How Many Demonstrations Is a Task Description Worth? Specificity and Reliability in In-Context Learning

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Abstract

Task descriptions and demonstrations both convey information about a task, but their relative value is difficult to quantify. We define the predictive effective sample size (ESS) of a description as the number of additional demonstrations a learner needs to match the reduction in prediction error that the description gives it. In a Bayesian model of in-context linear regression, we separate a description's specificity, the precision of its task information when relevant, from its reliability, the probability that it is relevant: a reliable description can replace many demonstrations, with a value that grows with the task's dimension as well as its precision, whereas any fixed probability of irrelevance caps the value of even an arbitrarily precise description without demonstrations. Transformers meta-trained on this task approach the Bayesian ESS and learn from their training distribution how much to trust descriptions. Pretrained language models given explicit probabilistic descriptions underweight precise descriptions when reasoning is off. With reasoning, most approach a Bayesian reference built from the prompt, and two models we retest, told what follows if a description is wrong, update their trust in it from the demonstrations, in numerical and semantic tasks. These behaviors depend on how explicitly the prompt states its statistical model.

1. Introduction

An in-context learner can learn about a task in two ways: a prompt can describe what to do, or it can demonstrate the desired behavior. Few-shot prompts often combine the two (Brown et al., 2020), and task descriptions also appear in system instructions and the files used to configure coding agents (Chatlatanagulchai et al., 2025). Yet we lack a common scale for comparing the information supplied by a description with that supplied by examples. When should a precise description replace many demonstrations? When can examples help a learner recognize that a description does not apply?

Empirical findings suggest that the answers depend on both the description and the learner. Prompts can substitute for substantial labeled data in prompt-based fine-tuning (Le Scao & Rush, 2021), and written summaries can replace demonstrations (Honda et al., 2025). Other studies find weak sensitivity to instruction relevance (Webson & Pavlick, 2022), reliance on demonstrations over explicit instructions (Gupta et al., 2025), or limited benefits from repository instruction files (Gloaguen et al., 2026). These settings differ, but they raise a shared measurement problem: how much task information does a description provide, and how much of it does the model use?

We address this problem within the Bayesian view of in-context learning (ICL), which treats the prompt as evidence about a latent task (Xie et al., 2022). A description induces a distribution over tasks before the demonstrations are observed. We characterize this distribution through two properties. *Specificity* measures how tightly a relevant description constrains the task. *Reliability* measures the probability that the description is relevant in the first place. A broad but correct description and a precise description of the wrong task create different inference problems: the latter requires the learner to assess whether the demonstrations conflict with the description (Evans & Moshonov, 2006).

To compare descriptions with demonstrations, we adapt the effective sample size of an informative prior (Morita et al., 2008; Wiesenfarth & Calderazzo, 2020; Reimherr et al., 2021). Our *predictive ESS* is the number of additional demonstrations that a learner without the description needs to match its own error with the description (Figure 1). The definition applies at any number of demonstrations and to any predictor. It therefore distinguishes the benefit available to a Bayesian learner from the benefit realized by a trained model.

We develop this framework in Gaussian linear regression,

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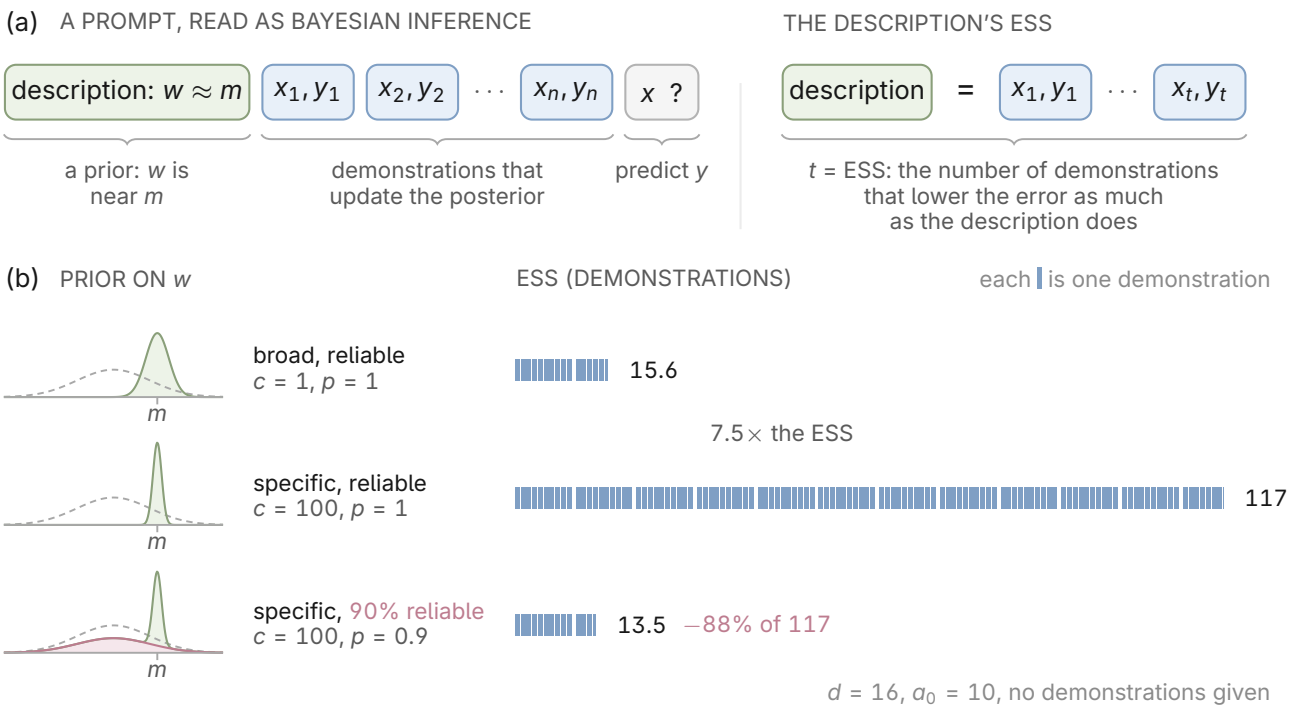


Figure 1. **Measuring a task description in demonstrations.** (a) A description provides information about a latent task w; demonstrations supply additional evidence. Its effective sample size (ESS) is the number of additional demonstrations needed to achieve the same prediction error. (b) Baseline and description-conditioned task distributions, shown schematically, with their Bayesian ESS at $n = 0$, $d = 16$, and $a_0 = 10$. Specificity c is the precision of a relevant description; reliability p is its probability of relevance. Pink denotes an irrelevant description. Increasing specificity can greatly increase ESS when $p = 1$, whereas a 10% probability of irrelevance sharply limits this benefit without demonstrations.

where description information and prediction error can be analyzed directly. We then study two ways a learner can acquire that information. Meta-trained Transformers learn the specificity and reliability of description tokens from their training distribution. Pretrained language models instead receive explicit probabilistic descriptions in text. This second setting isolates whether models use stated uncertainty when combining a description with demonstrations. Its scalar task tests precision and reliability but not the dimension effects, which we test with the meta-trained Transformers.

Our contributions are threefold:

- **A predictive measure of description value.** We adapt risk-matching ESS to task descriptions, using each learner's own learning curve as the baseline (Definition 1).
- **An analysis of specificity and reliability.** We derive risk formulas and ESS limits for Gaussian task descriptions. Reliable descriptions benefit from both precision and coverage of task dimensions. Unreliability imposes a finite ceiling without demonstrations, while demonstrations can help assess relevance (Sections 3–4).
- **Tests of learned and stated task information.** Meta-trained Transformers approach the Bayesian benchmark. In pretrained language models, reasoning substantially improves the use of stated precision, and the two models retested with a stated fallback update relevance from the demonstrations, in numerical and semantic tasks (Sections 5–6).

2. Framework

Tasks and demonstrations. We use the standard noisy in-context linear regression setting (Garg et al., 2022; Raventós et al., 2023). A task is a vector $w \in \mathbb{R}^d$ drawn from the baseline prior $\mathcal{N}(0, a_0 I_d)$. A prompt contains n demonstrations $D_n = \{(x_i, y_i)\}_{i=1}^n$ and a query x, where

$$x_i, x \overset{\text{iid}}{\sim} \mathcal{N}(0, I_d), \qquad y_i = w^\top x_i + \epsilon_i, \quad \epsilon_i \overset{\text{iid}}{\sim} \mathcal{N}(0, 1).$$

The parameter $a_0 > 0$ controls variation across tasks. Numerical illustrations in Sections 3–4 use $d = 16$ and $a_0 = 10$.

Descriptions as uncertain task information. A description supplies a center $m \in \mathbb{R}^d$. Its *specificity* is a precision $c \geq 1/a_0$, and its *reliability* is a probability $p \in [0, 1]$. Write $r = (ca_0)^{-1} \in (0, 1]$ for the fraction of baseline variance remaining when the description is relevant. With an unob-

served relevance indicator $Z \sim \text{Bernoulli}(p)$, independent of m, we generate

$$m \sim \mathcal{N}(0, (1-r)a_0 I_d),$$
$$w \mid m, Z = 1 \sim \mathcal{N}(m, r a_0 I_d),$$
$$w \mid m, Z = 0 \sim \mathcal{N}(0, a_0 I_d).$$

Thus the description induces the mixture prior

$$w \mid m \sim p\,\mathcal{N}(m, c^{-1} I_d) + (1-p)\,\mathcal{N}(0, a_0 I_d),$$

as in robust mixture priors for potentially incompatible information (Schmidli et al., 2014). Both relevance branches have the same marginal task distribution, $w \sim \mathcal{N}(0, a_0 I_d)$. Consequently, the comparison with and without a description does not change the task population.

A description is *reliable* when $p = 1$ and *unreliable* when $p < 1$. The model separates uncertainty within the described region from uncertainty about whether that region applies; it does not assume that every natural-language description reduces to these two scalars.

Prediction error. For a prediction $\hat{y}$ of the query response, we measure error against its conditional mean:

$$R = \mathbb{E}\big[(\hat{y} - w^\top x)^2\big]. \tag{1}$$

This excludes irreducible query noise. The expectation is over tasks, descriptions, demonstrations, and queries. For a fixed learner, let $R^{\text{desc}}(n)$ and $R^{\text{demo}}(n)$ denote its errors with and without a description. We write $R^{\text{desc}}_{p=1}(n)$ for the reliable case.

Definition 1 (Predictive effective sample size)**.** Given n demonstrations, let n^* be the smallest $t \geq 0$ such that $R^{\text{demo}}(t) = R^{\text{desc}}(n)$. When this crossing exists, the ESS of the description is

$$\text{ESS}_n = n^* - n.$$

We interpolate R^{demo} between available demonstration counts: linearly in t for the theory and meta-trained models, and in $\log t$ for language models. For measured curves, we use the first crossing within the observed range and report out-of-range values as bounds. The language-model conventions are detailed in Appendix C. If $R^{\text{desc}}(n) > R^{\text{demo}}(t)$ for all $t \geq 0$, we report $\text{ESS}_n \leq -n$.

ESS measures a horizontal displacement of a learning curve: $\text{ESS}_n = 10$ means that n demonstrations plus the description match the error of $n+10$ demonstrations without it. For a decreasing baseline curve, a negative value indicates increased error at the same demonstration count. Each model is evaluated against its own curve, so ESS measures the description's value *to that learner*; a large ESS alone does not imply low absolute error. Under the correctly specified Bayesian model, descriptions cannot increase expected risk, and ESS is nonnegative.

3. The Value of a Reliable Description

For $p = 1$, the Bayesian predictor uses the posterior mean of $w^\top x$. Its risk is the expected trace of the posterior covariance of w. Let X_n contain the demonstration inputs as rows and write $W_n = X_n^\top X_n$. Then

$$R^{\text{demo}}(n) = \mathbb{E}\,\text{tr}(a_0^{-1} I_d + W_n)^{-1},$$
$$R^{\text{desc}}_{p=1}(n) = \mathbb{E}\,\text{tr}(c I_d + W_n)^{-1}.$$

The description replaces baseline precision a_0^{-1} with c. Its predictive value is not, however, determined by this precision difference alone.

Proposition 1 (Reliable descriptions)**.** *Let $p = 1$.*

(i) With $r \in (0, 1]$ fixed, $\text{ESS}_0 \geq (1-r)d$, and $\text{ESS}_0 \to (1-r)d$ as $a_0 \to \infty$.
(ii) With positive integer c fixed, $\text{ESS}_0 \to c + d + 1$ as $a_0 \to \infty$.

The two limits describe different regimes (proof in Appendix A.1). In the first, the description removes a fixed fraction of uncertainty in every direction. Each of the first d demonstrations constrains one direction, so reducing variance across all directions can save approximately $(1-r)d$ demonstrations. In the second, the description retains a fixed absolute precision while the baseline prior becomes diffuse. Without a description, the limiting risk is $d/(n-d-1)$ for $n > d+1$ (Muirhead, 1982); the description alone has risk d/c. These risks match at $n = c + d + 1$, a demonstration count when c is an integer.

This $d+1$ contribution reflects the cost of learning a d-dimensional task from random inputs. Matching average information would count only the added precision $c - a_0^{-1}$; matching predictive error also accounts for the random design. We refer to the additional contribution as the *dimension term*. It is substantial when c is small relative to d, whereas precision dominates when c is large.

At $d = 16$ and $a_0 = 10$, a reliable description with $c = 1$ has ESS 15.6, compared with the lower bound $(1-r)d = 14.4$. Increasing c to 100 raises ESS to approximately 117, matching $c + d + 1$ (Figure 2). As demonstrations accumulate, the dimension benefit disappears: $\text{ESS}_n \to c - a_0^{-1}$ (Corollary 3). For example, after 100 demonstrations the $c = 1$ description is worth 1.1 demonstrations, close to its added precision of 0.9.

Coverage matters as well as precision. A description may constrain only $k < d$ directions. For any fixed precision $c > 0$, its zero-demonstration ESS approaches k as $a_0 \to \infty$; increasing that fixed precision does not remove uncertainty in the other directions. With many demonstrations, its ESS instead approaches $k(c - a_0^{-1})/d$ (Proposition 4, Appendix A.3). A precise partial description can therefore

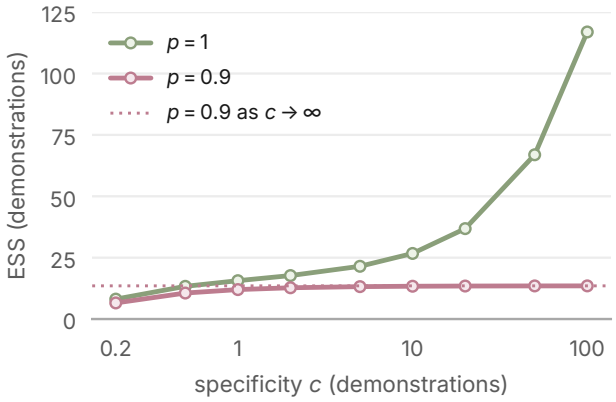


Figure 2. **Reliability limits the benefit of greater specificity without demonstrations.** Bayesian ESS_0 versus specificity c, for $p = 1$ and $p = 0.9$, at $d = 16$ and $a_0 = 10$. The dotted line is the finite ceiling for $p = 0.9$ as $c \to \infty$. Estimates use eight batches of 4,000 prompts; standard errors are below 0.02 demonstrations.

become more valuable as demonstrations resolve what it leaves unspecified. At $d = 16$, $k = 8$, and $c = 100$, ESS increases from 8.1 without demonstrations to 36 with 100 demonstrations (Table 1).

4. The Cost of Unreliability

When $p < 1$, the learner must also infer whether the description applies. Let μ_1 and μ_0 denote the posterior means conditional on relevance and irrelevance, and let $\pi_n = P(Z = 1 \mid m, D_n)$ be the posterior relevance probability. The Bayesian prediction is

$$\hat{y} = \big[\pi_n \mu_1 + (1 - \pi_n)\mu_0\big]^\top x.$$

Demonstrations affect both the component predictions and their relative weight.

Proposition 2 (Risk under uncertain relevance)**.** *For every $n \geq 0$,*

$$R^{\mathrm{desc}}(n) = pR^{\mathrm{desc}}_{p=1}(n) + (1 - p)R^{\mathrm{demo}}(n) + \mathcal{V}_n, \tag{2}$$

where

$$\mathcal{V}_n = \mathbb{E}\Big[\pi_n(1 - \pi_n)\big((\mu_1 - \mu_0)^\top x\big)^2\Big] \geq 0.$$

With no demonstrations,

$$R^{\mathrm{desc}}(0) = \big[1 - p^2(1 - r)\big]a_0 d. \tag{3}$$

Equation (2) separates the error of an oracle that knows Z from the cost of uncertainty about relevance. The oracle uses the description when it applies and otherwise uses the baseline prior. The remaining term, $\mathcal{V}_n$, is the disagreement between these two predictions, weighted by uncertainty about which is appropriate. Without demonstrations, the oracle removes a fraction $p(1 - r)$ of baseline risk; a learner that must infer relevance removes only $p^2(1 - r)$. The proof follows from the law of total variance (Appendix A.2).

Corollary 3 (ESS under uncertain relevance)**.**

(i) For $0 < p < 1$, define $n_p^\sharp$ by

$$R^{\mathrm{demo}}(n_p^\sharp) = (1 - p^2)R^{\mathrm{demo}}(0).$$

Then $\mathrm{ESS}_0 < n_p^\sharp$ at every finite c, and $\mathrm{ESS}_0 \to n_p^\sharp$ as $c \to \infty$. As $a_0 \to \infty$ with r fixed, $\mathrm{ESS}_0 \to p^2(1 - r)d$ and $n_p^\sharp \to p^2 d$. For $p = 0$, both ESS and the ceiling are zero.

(ii) With d, a_0, and c fixed, as $n \to \infty$,

$$\begin{cases} \mathrm{ESS}_n \to c - a_0^{-1}, & p = 1, \\ \limsup_{n\to\infty} \mathrm{ESS}_n \leq p(c - a_0^{-1}), & p < 1. \end{cases}$$

Specificity cannot remove a relevance error floor. For fixed $p < 1$, Equation (3) gives a positive risk floor $(1 - p^2)a_0 d$, even as $c \to \infty$. Hence zero-demonstration ESS has a finite ceiling. At $p = 0.9$, $d = 16$, and $a_0 = 10$, increasing c from 1 to 100 raises ESS only from 12.0 to 13.5, compared with 15.6 to 117 when $p = 1$ (Figure 2). The finite-a_0 ceiling is approximately 13.5; the diffuse-baseline approximation is $p^2 d = 12.96$.

Demonstrations distinguish low precision from uncertain relevance. Without demonstrations, an unreliable description has the same risk as a reliable description that leaves variance fraction $1 - p^2(1 - r)$. For example, $c = 100$, $p = 0.9$ and a reliable description with $c \approx 0.52$ both have ESS approximately 13.5 in this setting. Their values diverge once demonstrations are available: after 100 demonstrations, the former has ESS 86.6 and the latter 0.5. Evidence can support the high-precision component of the unreliable description; the reliable low-precision description has no such component to recover.

Treating every description as relevant can be harmful. A fully trusting predictor always uses μ_1, even when the description is irrelevant. Without demonstrations, it is worse than ignoring the description when $p < 1/2$, provided the description is informative ($r < 1$). For fixed n, its risk tends to $2(1 - p)a_0 d$ as $c \to \infty$, whereas the demonstrations-only risk tends to zero as n grows. Thus a sufficiently precise, occasionally irrelevant description can outweigh useful demonstrations. At $p = 0.9$ and $c = 100$, full trust is worse than ignoring the description from 15 demonstrations onward (Appendix A.4).

5. Meta-Trained Transformers

We first test whether Transformers can learn to use descriptions in the setting where the Bayesian model is correctly specified. Following Garg et al. (2022), we train from scratch on linear regression tasks with $d = 5$ and $a_0 = 10$. A prompt contains an optional description token carrying m, up to 20 demonstration tokens, and a query token (Huang & Ge, 2025). The token does not specify c or p; the model must infer these properties from its training distribution. Half the training prompts omit the description, providing a demonstrations-only learning curve for each network. We train separate networks for $c \in \{1, 10\}$ and $p \in \{1, 0.99, 0.9\}$, with two or three seeds per configuration. Architecture, optimization, and additional experiments are reported in Appendix B.

Matched training and test distributions. With no demonstrations, the $c = 10$, $p = 1$ networks achieve mean ESS 14.3 (Figure 3a), compared with a Bayesian ESS of 15.9 on a larger sample (Table 2). At lower reliability, and for $c = 1$, the zero-demonstration ESS is within 0.3 of the Bayesian value. The networks also capture the increased value of a partial description as demonstrations accumulate: with $k = 3$ of $d = 5$ directions described at $c = 10$, ESS rises from 3.27 at $n = 0$ to 5.55 at $n = 10$, compared with Bayesian values of 3.22 and 5.64.

Reliability shifts. We also evaluate each network at reliabilities different from its training value. We compare it with a *train-matched* Bayesian predictor, which retains the training reliability, and a *calibrated* predictor, which uses the test reliability (Raventós et al., 2023). At $n = 0$, the Bayesian prediction is $p\, m^\top x$. We therefore estimate a network's *implied trust* as the coefficient q that best approximates its predictions by $q\, m^\top x$, following coefficient-based analyses of learned predictors (Akyürek et al., 2023; Panwar et al., 2024).

For the $c = 10$ description, networks trained at $p = 0.9$ retain implied trust near 0.9 even when every test description is relevant, falling short of the calibrated predictor (Figure 3b, Table 2). This is consistent with the information available: without demonstrations, m itself does not reveal the test reliability. Networks trained at $p = 1$ instead continue to trust descriptions fully. When 10% of test descriptions are irrelevant, their prediction error is higher with the description than without it from six demonstrations onward, in every seed. Training at $p = 0.99$ avoids this failure in the tested range: descriptions continue to reduce error at every evaluated n. Exposure to occasional irrelevance during training therefore changes how these networks combine descriptions with later evidence.

6. Pretrained Language Models

The meta-trained networks learn description statistics from repeated tasks. We next ask whether pretrained language models use the same information when it is stated explicitly. We use numerical mean estimation, a scalar version of the regression problem with fixed input $x = 1$, following controlled tests of stated priors (Gupta et al., 2025), and then a semantic classification task. Both tasks state the information that defines the reference, so a gap from it reflects how a model uses that information.

6.1. Task and Evaluation

Each task has a hidden mean $w \sim \mathcal{N}(500, 100^2)$ and demonstrations drawn from $\mathcal{N}(w, \sigma^2)$, with $\sigma = 60$, then rounded to integers. A description supplies a value m and uncertainty τ, for example: "Its mean was drawn from a normal distribution with mean 510 and standard deviation 30." The underlying Gaussian construction has $w \mid m \sim \mathcal{N}(m, \tau^2)$. We express precision in demonstration units as $c = \sigma^2/\tau^2$: stated standard deviations 60, 30, and 10 correspond to $c = 1$, 4, and 36.

We vary the stated precision, and add conditions stating a 50% probability that the description is wrong. In these conditions the description remains relevant; the intervention tests how models respond to the stated probability. In a fixed-value control, we hold m and the demonstrations fixed while describing a value generated at $c = 36$ as having $c = 4$. This separates the response to the stated uncertainty from the response to the value itself. Appendix C includes genuinely irrelevant descriptions and evaluations under actual mixtures of relevant and irrelevant descriptions.

The main comparison uses the same 200 tasks for five models with reasoning enabled and disabled: gpt-oss-20b (OpenAI, 2025), Qwen3.5 9B (Qwen Team, 2026), Gemma 4 31B (Gemma Team, 2026), GPT-6 Sol (OpenAI, 2026), and Claude Opus 4.6 (Anthropic, 2026b); Claude Haiku 5.5 (Anthropic, 2026a) is evaluated with reasoning at medium effort. Reasoning is each model's own mode for generating reasoning before it answers, switched on or off through its provider's setting. Each pair uses the same model identifier, but for the open-weight models the two modes also differ in serving software and decoding. Appendix C.4 reports these details, effort settings, response extraction, and reruns, and Appendix C a larger evaluation without reasoning on 2,000 tasks.

A reference based on the prompt. The prompt specifies the observation noise and description, but not the task generator's baseline prior. Our *prompt-based Bayesian reference* therefore uses the stated description and, for a wrong description, a uniform fallback over the three-digit range. For a description asserted to be relevant, ignoring negligible

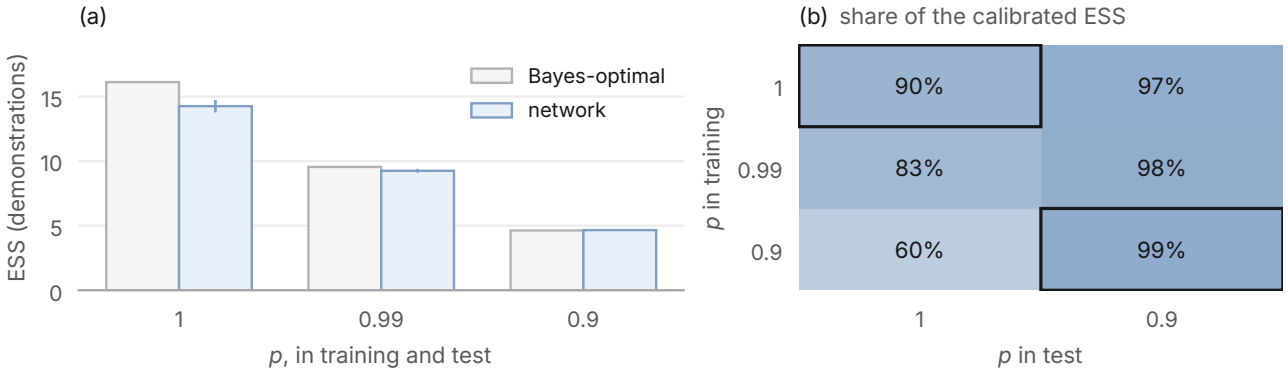


Figure 3. **Meta-trained Transformers approach the Bayesian ESS and inherit their trust from training.** Results for $c = 10$, $d = 5$, $a_0 = 10$, and no demonstrations. (a) Training and test reliability match. Network ESS uses each network's own demonstrations-only curve; bars show the mean and one standard deviation across three training seeds. The Bayesian comparison uses the same evaluation prompts. (b) Reliability changes at test time. Cells show network ESS as a fraction of the calibrated Bayesian ESS; outlined cells have matching training and test reliability. Table 2 also reports the $c = 1$ condition.

boundary effects, its prediction is

$$\hat{w}_{\text{ref}} = \frac{c\,m + n\bar{y}}{c + n}, \tag{4}$$

where $\bar{y}$ is the demonstration mean. With a fixed input there is no random design, so under a flat baseline prior the description has exact ESS c.

This reference follows the prompt even where a stated precision or reliability differs from the data generator, so it models inference from the stated information rather than optimal prediction. When the description may be wrong, it also depends on the assumed fallback (Appendix C).

For each model, we compute ESS from its own demonstrations-only curve and summarize it by the median over $n \in \{1, 2, 4\}$. The same calculation gives reference ESS 34.6 for $c = 36$, rather than the idealized value 36, because of finite-sample error and interpolation. We also regress predictions on m, $\bar{y}$, and an intercept. The coefficient on m measures reliance on the description; the reference coefficient is $c/(c + n)$ when the description is asserted to be relevant.

6.2. Reasoning and Stated Precision

Underweighting without reasoning. For the $c = 36$ description, ESS without reasoning ranges from 2.0 to 10.3 across the five models, compared with 34.6 for the reference (Figure 4a): every model gives the precise description far less weight than its stated precision warrants. Their response to the stated precision varies. When the same value is described as $c = 4$ instead of $c = 36$, reference ESS falls from 34.6 to 13.7; GPT-6 Sol's falls from 10.3 to 2.4, Opus 4.6's and Gemma 4 31B's fall by about half, and those of Qwen3.5 9B and gpt-oss-20b change less (Figure 4b). At $n = 4$, GPT-6 Sol and Opus 4.6 weight the value 0.47 and 0.40 at stated $c = 36$ and 0.19 and 0.20 at stated $c = 4$, against reference weights 0.90 and 0.50.

With reasoning, several models weight precision as the reference does. With reasoning enabled, Opus 4.6 and Gemma 4 31B reach approximately the reference ESS in both precision conditions, and GPT-6 Sol comes close. Haiku 5.5 at medium effort falls short (28 and 8.5, against 34.6 and 13.7), mainly at $n = 1$, where it skips reasoning in 21 to 28% of replies. On tasks where m and $\bar{y}$ differ enough for one reply to identify a weight, the weight implied by an individual reply at $n = 4$ is within 0.05 of the reference weight in 89–100% of these four models' replies (Appendix C.5). A capability benchmark does not explain this: gpt-oss-20b with reasoning reaches the reference's weights, whereas Opus 4.6 without it does not, although Opus 4.6 scores 19.1% on Humanity's Last Exam (HLE) against 5.3%.

Qwen3.5 9B and gpt-oss-20b also approach the reference precision weights. Their demonstrations-only curves flatten at larger n, for gpt-oss-20b through arithmetic mistakes, so some crossings lie beyond the grid. For these cases, square markers in Figure 4 use the reference's demonstrations-only curve instead, a different quantity from learner-specific ESS. Olmo 3 7B Think, another model with a reasoning mode, averages demonstrations well but often returns the description value alone (ESS 6.2 at $c = 36$; Appendix C.5), so reasoning mode is not sufficient.

6.3. Stated Reliability and Wording

With a stated fallback, the retested models update relevance. Adding a stated 50% probability of error lowers reference ESS from 34.6 to 5.7 under the uniform fallback. Without reasoning, models' ESS falls much less than this

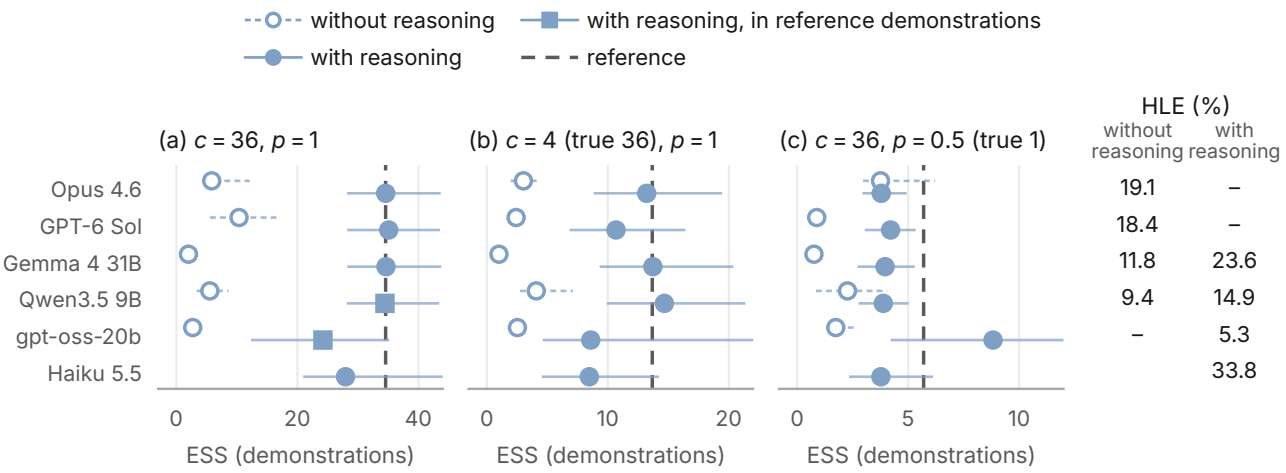


Figure 4. **Reasoning improves the use of stated precision, with model-specific exceptions.** Summary ESS on 200 tasks at $\sigma = 60$ (median over $n = 1, 2, 4$), with paired 95% bootstrap intervals over tasks: open points without reasoning, filled with it; vertical dashed lines, the prompt-based reference. (a) Relevant description with $c = 36$. (b) The same value stated to have $c = 4$; reference ESS is 13.7, not 4, because the value is more accurate than stated. (c) The description in (a), stated to have a 50% chance of being wrong although it is relevant; reference ESS 5.7 under its assumed uniform fallback. Squares for Qwen3.5 9B and gpt-oss-20b in (a) use the reference's demonstrations-only curve, because the models' own curves do not reach the crossing. Intervals are clipped to the displayed range; three have no upper end on the grid. HLE: Humanity's Last Exam scores (Center for AI Safety et al., 2026) reported by Artificial Analysis (2026); dashes mark unavailable scores, blanks modes not evaluated.

drop, except GPT-6 Sol's, from 10.3 to 0.9. With reasoning, GPT-6 Sol, Gemma 4 31B, and Qwen3.5 9B closely track a different rule: they retain the stated relevance probability and weight m by $pc/(c+n)$ (Table 9). This averages the relevant and irrelevant predictions using the *prior* mixture weights.

Bayesian inference instead updates relevance by comparing the likelihood of the demonstrations with and without the description. The prompt says only that the mean is then "unknown", which leaves the second likelihood undefined, so keeping the stated probability is defensible. We therefore rerun the condition for two models, GPT-6 Sol at low effort and Haiku 5.5 at medium effort, with the warning instead saying that the mean "was instead drawn uniformly at random from the whole numbers 100 to 999", the reference's fallback. On tasks where the reference and fixed-rule predictions differ by at least 5, the share of replies closer to the reference rises from 0–5% to 92–97% for GPT-6 Sol and from 21–41% to 87–95% for Haiku 5.5. Each reply's implied trust in the description then tracks the reference's posterior relevance task by task, with regression slopes 0.96 and 0.98, against 0.01 and 0.22 under the original warning (Appendix C.5). At the stated 50%, both models then update relevance from the demonstrations about as the reference does.

More explicit statistical descriptions are used more consistently. We test three alternative wordings with GPT-6 Sol and Haiku 5.5. Both closely follow Equation (4) under the original wording and "An earlier estimate put its mean at m, with a standard error of τ," except Haiku 5.5 at $n = 1$. Under "Its mean is about m, give or take τ" and "Mean: $m \pm \tau$," they often return m and disregard demonstrations. With the first informal wording, GPT-6 Sol skips reasoning in 24–51% of replies per condition and usually returns m in those replies. Haiku 5.5 returns m in 71% of the replies for which it does reason.

The informal wordings keep the numbers but do not say that τ is a standard deviation or how the mean was generated. As with the unstated fallback, we therefore read this as an effect of how explicitly the prompt states its statistical model, not as a failure to treat equivalent phrasings alike. GPT-6 Sol's wording runs used an earlier agent configuration that also changed how often it reasons (Appendix C.4).

6.4. A Semantic Task

With reasoning, two models use semantic descriptions almost as the reference does. In our semantic task, a company files messages into one of two folders by a hidden policy: each message states six properties in ordinary language, and the policy requires two of them to be present or absent. The prompt states that each of the 60 such policies is equally likely, so the Bayesian reference is exact (Appendix C.7). A specific note states one requirement and names the other property. With reasoning, Haiku 5.5 and GPT-6 Sol value it at 9.9 and 9.5 demonstrations at $n = 2$, against 11.4 for the reference, with ESS from the squared

error of their stated probabilities. Given a note of this form with a stated 50% chance of being wrong, which it is, and the fallback stated, their predictions follow the reference's update rather than the stated chance on 77 to 99% of the messages where the two differ. Without reasoning, GPT-6 Sol's loss with the specific note at $n = 2$ is 0.130, against 0.027 for the reference.

7. Related Work

Bayesian accounts of ICL. ICL can be viewed as inference about a latent task (Xie et al., 2022), and Transformers trained on task distributions can approximate posterior prediction (Müller et al., 2022; Akyürek et al., 2023; von Oswald et al., 2023), though they can depart from it under low task diversity or for new function classes (Raventós et al., 2023; Panwar et al., 2024), and in-context learning in language models violates properties of Bayesian updating (Falck et al., 2024); we use Bayesian predictors as references, not as models of the learner. Work incorporating additional task information considers input-distribution descriptions (Huang & Ge, 2025), candidate hypotheses (Lin et al., 2025), mixtures of task groups (Lin & Lee, 2024), how an instruction's influence decays with demonstrations (Tong et al., 2026), and prefixes of related datasets for hierarchical Bayesian prediction (Zhu et al., 2026). We focus on a common quantitative question across such settings: how much does task information reduce the number of demonstrations needed for a given prediction error?

Prior effective sample size. ESS is an established way to quantify informative priors (Morita et al., 2008; Neuenschwander et al., 2020). For a Bayesian predictor, our measure is an expected-error form of the effective current sample size (Wiesenfarth & Calderazzo, 2020), which builds on risk-based definitions (Reimherr et al., 2021); for other learners, it is the horizontal gap between learning curves that Le Scao & Rush (2021) call the average data advantage. Our contribution is to compare the value a description offers a Bayesian learner with the value a trained learner realizes, and to analyze how specificity and reliability set the former. Related work discounts incompatible historical information using predictive loss (Reznik, 2026); robust mixture priors (Schmidli et al., 2014) and discounting of possibly irrelevant sources (Pichon et al., 2012) similarly account for possible mismatch. Selby et al. (2025) evaluate priors elicited from language models by their predictive sample value. We instead measure descriptions supplied to an in-context learner.

Descriptions and evidence in language models. Controlled studies find that demonstrations can outweigh stated priors (Gupta et al., 2025). Other work examines semantic priors and flipped labels (Wei et al., 2023), weak dependence on demonstration labels (Min et al., 2022), Bayesian scaling laws (Arora et al., 2025), and Bayesian reference curves for in-context function learning (Akata et al., 2026). Concurrent work finds that language models violate Bayes' rule in context but closely approximate posteriors with reasoning and code execution (Filbry et al., 2026). In our main runs we disable code execution, vary stated precision and reliability separately, use fixed-value controls, and measure the result in demonstrations. Appendix D reviews more work.

8. Discussion

Description ESS connects two questions: how much task information is available, and how much of it a learner uses. In our model, a reliable description can replace many demonstrations because it constrains the task across directions as well as increasing precision. Unreliability changes this relationship: without demonstrations, even an arbitrarily precise description has bounded value. Demonstrations can then serve a second purpose, providing evidence about whether the description applies. Precise partial descriptions also gain value as demonstrations resolve what they leave unspecified.

The empirical results show that these benefits are learnable but not automatic. Meta-trained Transformers approach the Bayesian benchmark under matched task distributions, while their responses to reliability shifts reflect training. With reasoning enabled, most pretrained language models we test weight stated precision almost as the reference does, and the two models given a stated fallback update relevance from the demonstrations, in numerical and semantic tasks. When the prompt leaves the fallback open, several keep the stated probability fixed, and when it states uncertainty informally, they often ignore the demonstrations. These findings characterize prediction behavior; they do not identify an internal mechanism, and our design does not separate reasoning from the arithmetic it makes possible.

Scope and limitations. The theory assumes Gaussian linear regression, squared error, and one mixture model of irrelevance. Because ESS uses each learner's own curve, it compares how much learners gain from a description, not how well they predict. The language-model tasks are small and state their probabilistic information; natural descriptions without it, scored against the task generator, would test whether models infer it. The main comparison uses 200 tasks, and the wording and stated-fallback tests cover two models. Mode comparisons can differ in serving settings, and GPT-6 Sol's ESS without reasoning depends on its agent configuration. Open-ended constraints, structured tasks, and heavy-tailed description priors (Dalal & Hall, 1983; O'Hagan & Pericchi, 2012) would test how far these relationships extend.

Impact Statement

This work studies how learners combine task descriptions with demonstrations. Its measurement framework may inform the evaluation and design of prompts, especially when descriptions are uncertain or incomplete. The experiments use synthetic prediction tasks. Conclusions about these tasks should not be taken as evidence of reliable instruction following in consequential applications; the broader implications depend on how the framework is extended and used.

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A. Proofs

A.1. Reliable Descriptions

Notation and risk. Write $\varepsilon = 1/a_0$, let $X_n \in \mathbb{R}^{n\times d}$ contain the demonstration inputs as rows, and set $W_n = X_n^\top X_n$. For $\kappa > 0$, define

$$R^\kappa(n) = \mathbb{E}\,\mathrm{tr}(\kappa I_d + W_n)^{-1}.$$

Thus $R^{\mathrm{demo}} = R^\varepsilon$ and $R^{\mathrm{desc}}_{p=1} = R^c$. These identities follow from the posterior covariance under Gaussian linear regression. Conditional on the prompt, the posterior mean minimizes squared prediction error, and the independent query satisfies $\mathbb{E}[xx^\top] = I_d$. Averaging its conditional error $x^\top \Sigma x$ therefore gives $\mathbb{E}\,\mathrm{tr}\,\Sigma$.

Throughout the proofs, d is a positive integer, $a_0 > 0$, $c \geq \varepsilon$, $r = 1/(ca_0) \in (0, 1]$, and $p \in [0, 1]$. Noninteger sample sizes refer to linear interpolation between the risks at adjacent integers.

Wishart identities. We use three standard properties of W_n.

(W1) If $n < d$, then W_n has rank n almost surely. Consequently, $(\kappa I_d + W_n)^{-1}$ has $d - n$ eigenvalues equal to $1/\kappa$.

(W2) If $n \geq d$ and e is a fixed unit vector, then $(e^\top W_n^{-1} e)^{-1} \sim \chi^2_{n-d+1}$ (Muirhead, 1982, Theorem 3.2.12). Hence

$$\mathbb{E}\,\mathrm{tr}\,W_n^{-1} = \frac{d}{n-d-1} \quad (n \geq d+2),$$

whereas this expectation is infinite for $n = d$ and $n = d + 1$.

(W3) For fixed d, as $n \to \infty$,

$$\mathbb{E}\,\mathrm{tr}\,W_n^{-2} = \frac{d}{n^2} + O(n^{-3}), \qquad \mathbb{E}\,\mathrm{tr}\,W_n^{-3} = O(n^{-3})$$

(von Rosen, 1988).

The Sherman–Morrison identity implies that an additional demonstration strictly decreases the expected trace of the posterior covariance. Thus $R^\kappa(n)$ is strictly decreasing, and (W2) implies that it tends to zero. Its linear interpolation is therefore continuous and strictly decreasing. In the correctly specified joint model, conditioning on the description cannot increase expected posterior variance, so $R^{\mathrm{desc}}(n) \leq R^{\mathrm{demo}}(n)$ and the Bayes-optimal ESS is nonnegative.

Proof of Proposition 1. A reliable description without demonstrations has risk

$$R^{\mathrm{desc}}_{p=1}(0) = \frac{d}{c} = ra_0 d.$$

(i) Fixed variance reduction. By (W1), $R^\varepsilon(n) \geq (d-n)a_0$ for $n < d$; the inequality also holds trivially for $n \geq d$. Linear interpolation preserves this bound, so

$$R^{\mathrm{demo}}(t) \geq (d-t)a_0 \qquad (t \geq 0).$$

For $t < (1-r)d$, the right-hand side exceeds $ra_0 d$. The matching sample size must therefore satisfy $\mathrm{ESS}_0 \geq (1-r)d$.

For every integer n,

$$\frac{R^\varepsilon(n)}{a_0} = \mathbb{E}\,\mathrm{tr}(I_d + a_0 W_n)^{-1} \longrightarrow (d-n)^+ \qquad \text{as } a_0 \to \infty.$$

Indeed, an eigenvalue $1/(1 + a_0\lambda)$ is bounded by one and converges to the indicator of $\lambda = 0$, so dominated convergence and the almost-sure rank of W_n apply. The same limit holds at every real t by interpolation. For fixed $r \in (0, 1)$, the limiting curve crosses rd at the unique point $t = (1-r)d$. Strict monotonicity then gives convergence of the crossing. For $r = 1$, the description is uninformative and $\mathrm{ESS}_0 = 0$ for every a_0, giving the same conclusion.

(ii) Fixed precision. For fixed $c > 0$, the description risk d/c is independent of a_0. As $a_0 \to \infty$, monotone convergence, (W1), and (W2) give

$$R^\varepsilon(n) \longrightarrow \begin{cases} +\infty, & n \leq d+1, \\ \dfrac{d}{n-d-1}, & n \geq d+2. \end{cases}$$

If c is a positive integer, the finite part of this limiting curve equals d/c at the integer $n = c + d + 1$. For every fixed point below this crossing the limiting risk is larger than d/c, and for every fixed point above it the limiting risk is smaller. The crossing of the interpolated finite-a_0 risk therefore converges to $c + d + 1$. The integer condition ensures that the equality occurs at a grid point; linear interpolation does not in general preserve the same expression for noninteger c. □

Large-sample expansion. For fixed d and κ, commuting W_n with κI_d gives

$$(\kappa I_d + W_n)^{-1} = W_n^{-1} - \kappa W_n^{-2} + \kappa^2 W_n^{-2}(\kappa I_d + W_n)^{-1}.$$

The final term is positive semidefinite and bounded above by $\kappa^2 W_n^{-3}$. Using (W2) and (W3),

$$R^{\kappa}(n) = \frac{d}{n-d-1} - \frac{\kappa d}{n^2} + O(n^{-3}). \tag{5}$$

An additional demonstration reduces the leading term by $d/n^2 + O(n^{-3})$. Thus a fixed change in prior precision and a fixed change in sample size affect risk at the same order. This observation underlies the large-n ESS limits below.

A.2. Unreliable Descriptions

Let $\mathcal{P}_n = (D_n, m, x)$ denote the full prompt, including the query input, and write $\pi_n = P(Z = 1 \mid \mathcal{P}_n)$. Conditional on relevance, the posterior parameters are

$$\Sigma_1 = (cI_d + W_n)^{-1}, \qquad \mu_1 = \Sigma_1(cm + X_n^\top y_{1:n}).$$

Conditional on irrelevance, m is independent of w, and

$$\Sigma_0 = (\varepsilon I_d + W_n)^{-1}, \qquad \mu_0 = \Sigma_0 X_n^\top y_{1:n}.$$

The relevance probability is obtained from the two Gaussian marginal likelihoods,

$$\begin{aligned} L_1 &= \mathcal{N}\big(y_{1:n}; X_n m, I_n + c^{-1} X_n X_n^\top\big)\,, \\ L_0 &= \mathcal{N}\big(y_{1:n}; 0, I_n + a_0 X_n X_n^\top\big)\,, \\ \pi_n &= \frac{pL_1}{pL_1 + (1-p)L_0}. \end{aligned}$$

The query does not enter this update because it is independent of Z. Without demonstrations, both marginal likelihoods equal one and $\pi_0 = p$: the distribution of m is the same under both hypotheses. At $p = 0$ or $p = 1$, the formula reduces to the corresponding single component.

Proof of Proposition 2. The posterior mean prediction is

$$\mathbb{E}[w^\top x \mid \mathcal{P}_n] = \big(\pi_n \mu_1 + (1-\pi_n)\mu_0\big)^\top x.$$

Applying the law of total variance over Z yields

$$\begin{aligned} \mathrm{Var}(w^\top x \mid \mathcal{P}_n) = \pi_n x^\top \Sigma_1 x + (1-\pi_n) x^\top \Sigma_0 x \\ + \pi_n(1-\pi_n)\big((\mu_1 - \mu_0)^\top x\big)^2. \end{aligned}$$

For any integrable function g of the prompt,

$$\mathbb{E}[\pi_n g(\mathcal{P}_n)] = \mathbb{E}[Z g(\mathcal{P}_n)] = p\,\mathbb{E}[g(\mathcal{P}_n) \mid Z = 1].$$

Conditional on $Z = 1$, the prompt follows the reliable-description model. Conditional on $Z = 0$, the description provides no information about the task. Taking expectations in the variance decomposition therefore gives

$$R^{\mathrm{desc}}(n) = pR^{\mathrm{desc}}_{p=1}(n) + (1-p)R^{\mathrm{demo}}(n) + \mathcal{V}_n,$$

with $\mathcal{V}_n$ as in Equation (2).

At $n = 0$, $\mu_1 = m$, $\mu_0 = 0$, and the prediction is $p\, m^\top x$. For each coordinate,

$$\mathrm{Var}(w_i) = a_0, \qquad \mathrm{Var}(m_i) = (1-r)a_0, \qquad \mathrm{Cov}(w_i, m_i) = p(1-r)a_0.$$

It follows that

$$\mathbb{E}(w_i - pm_i)^2 = a_0 - 2p^2(1-r)a_0 + p^2(1-r)a_0 = \big(1 - p^2(1-r)\big)a_0.$$

Since $\mathbb{E}[xx^\top] = I_d$, the prediction risk is $\mathbb{E}\|w - pm\|^2$. Summing over coordinates proves Equation (3). □

Proof of Corollary 3. (i) No demonstrations. For $0 < p < 1$ and finite c,

$$R^{\mathrm{desc}}(0) = \big(1 - p^2(1-r)\big)R^{\mathrm{demo}}(0) > (1-p^2)R^{\mathrm{demo}}(0).$$

The risk on the left decreases to the right-hand side as $c \to \infty$, equivalently $r \to 0$. The interpolated baseline risk is continuous, strictly decreasing, and tends to zero. Consequently, the matching sample size is strictly below the unique finite $n_p^\sharp$ defined in the corollary and converges to it as $c \to \infty$. At $p = 0$, the description is always irrelevant, so $\mathrm{ESS}_0 = n_0^\sharp = 0$; the non-strict bound remains valid.

The baseline lower bound and normalized limit established in Appendix A.1 also give

$$\mathrm{ESS}_0 \geq p^2(1-r)d, \qquad \mathrm{ESS}_0 \longrightarrow p^2(1-r)d$$

when $a_0 \to \infty$ with r fixed. Applying the same argument to the level $1 - p^2$ gives $n_p^\sharp \to p^2 d$. These are broad-prior limits, not exact identities at finite a_0.

For comparison, an oracle told Z has zero-demonstration risk

$$\big(1 - p(1-r)\big)a_0 d \geq (1-p)a_0 d.$$

Its limiting ceiling is therefore pd demonstrations as $a_0 \to \infty$. The additional reduction from p to p^2 in the non-oracle expression is the cost of uncertainty about relevance.

(ii) Many demonstrations. Fix d, a_0, and c, and define the oracle risk

$$L(n) = pR^c(n) + (1-p)R^\varepsilon(n).$$

Equation (5) implies, uniformly over integers $0 \leq t \leq T = \lceil c \rceil + 1$,

$$R^{\mathrm{demo}}(n+t) - L(n) = \frac{d}{n^2}\left[p(c-\varepsilon) - t + O(n^{-1})\right].$$

Linear interpolation gives the same expansion for real $t \in [0, T]$. Let $n + \eta_n$ be the sample size at which the baseline risk equals $L(n)$. Then

$$\eta_n = p(c-\varepsilon) + O(n^{-1}).$$

When $p(c-\varepsilon) = 0$, the two risks coincide and $\eta_n = 0$ exactly. If $p = 1$, the description risk equals $L(n)$, so

$$\mathrm{ESS}_n \longrightarrow c - \varepsilon = (1-r)c.$$

If $p < 1$, the variance decomposition gives $R^{\mathrm{desc}}(n) = L(n) + \mathcal{V}_n \geq L(n)$. Since the baseline risk decreases with sample size, $\mathrm{ESS}_n \leq \eta_n$, proving

$$\limsup_{n\to\infty} \mathrm{ESS}_n \leq p(c-\varepsilon) = p(1-r)c.$$

The nonnegative term $\mathcal{V}_n$ is why the result is an upper bound for unreliable descriptions. □

A.3. Partial Descriptions

A reliable description may constrain only part of a task. Suppose it leaves a fraction r of the baseline variance in $k < d$ directions and leaves the remaining $d-k$ directions unchanged. Its conditional prior precision is c in the first set of directions and $\varepsilon = 1/a_0$ in the second. Write

$$\Lambda = \operatorname{diag}(cI_k, \varepsilon I_{d-k}).$$

The prediction risk is $\mathbb{E}\operatorname{tr}(\Lambda + W_n)^{-1}$. Rotational invariance of W_n means that the result depends on the number of constrained directions, not on their orientation.

Proposition 4 (ESS of a partial description)**.** *Let $p = 1$ and let $0 \le k < d$ be an integer.*

(i) As $a_0 \to \infty$, $\mathrm{ESS}_0 \to (1-r)k$ with r fixed, and $\mathrm{ESS}_0 \to k$ with c fixed.
(ii) For fixed d, a_0, and c, as $n \to \infty$,

$$\mathrm{ESS}_n \longrightarrow \frac{k}{d}(c - 1/a_0) = \frac{k}{d}(1-r)c.$$

In the broad-prior limit, the unconstrained directions dominate the initial error. Even an arbitrarily precise description of $k < d$ directions therefore replaces only about k demonstrations when used alone. After many demonstrations, its value instead approaches the added precision averaged across all d directions. These two limits permit complementarity: a sufficiently precise partial description can be worth more alongside demonstrations than by itself. The numerical example in Table 1 shows this effect. At $c = 100$, a description of half of the $d = 16$ directions has ESS 8.1 without demonstrations and 36 with 100 demonstrations.

Proof of Proposition 4. (i) No demonstrations. The description risk is

$$\operatorname{tr}\Lambda^{-1} = (kr + d - k)a_0.$$

Using $R^{\mathrm{demo}}(t) \ge (d-t)a_0$ gives $\mathrm{ESS}_0 \ge (1-r)k$. The normalized baseline risk converges to $(d-t)^+$, locally uniformly in t, by the integer limits and linear interpolation. With r fixed, the normalized description risk is $d-(1-r)k$. With c fixed, it is

$$d - k + \frac{k}{ca_0} \longrightarrow d - k.$$

Because $k < d$, both limiting levels are positive and have unique crossings, at $(1-r)k$ and k, respectively. Convergence of the crossings proves the claim. When the description leaves the prior unchanged, its ESS is identically zero.

(ii) Many demonstrations. Applying the resolvent identity twice gives

$$(\Lambda + W_n)^{-1} = W_n^{-1} - W_n^{-1}\Lambda W_n^{-1} + W_n^{-1}\Lambda W_n^{-1}\Lambda(\Lambda + W_n)^{-1}.$$

Rotational invariance implies

$$\mathbb{E}W_n^{-2} = \frac{\mathbb{E}\operatorname{tr}W_n^{-2}}{d} I_d.$$

The expected trace of the middle term is therefore $\operatorname{tr}(\Lambda)/n^2 + O(n^{-3})$ by (W3). To bound the remainder, use $|\operatorname{tr}(MB)| \le \|M\|_{\mathrm{op}} \operatorname{tr} B$ for $B \succeq 0$, together with $(\Lambda + W_n)^{-1} \preceq W_n^{-1}$. Its absolute trace is at most

$$\|\Lambda\|_{\mathrm{op}}^2 \|W_n^{-1}\|_{\mathrm{op}}^2 \operatorname{tr} W_n^{-1} \le d\|\Lambda\|_{\mathrm{op}}^2 \operatorname{tr} W_n^{-3},$$

whose expectation is $O(n^{-3})$. Consequently,

$$\mathbb{E}\operatorname{tr}(\Lambda + W_n)^{-1} = \frac{d}{n-d-1} - \frac{\operatorname{tr}\Lambda}{n^2} + O(n^{-3}).$$

Matching this expansion to the baseline risk, as in Corollary 3(ii), gives

$$\mathrm{ESS}_n \longrightarrow \frac{\operatorname{tr}\Lambda - d\varepsilon}{d} = \frac{k(c-\varepsilon)}{d}.$$

□

The same expansion applies to any fixed Gaussian description precision $\Lambda \succeq \varepsilon I_d$ compatible with the baseline task distribution. Its large-sample ESS is the added precision averaged across directions, $(\operatorname{tr}\Lambda - d\varepsilon)/d$.

Table 1. **Demonstrations can increase the value of a precise partial description.** Bayes-optimal ESS for a reliable description of k of $d = 16$ directions, with $a_0 = 10$. The case $k = 16$ is a complete description. Each block reports ESS after n demonstrations and its large-sample limit $(1 - r)ck/d$. Estimates use eight batches of 2,000 prompts; standard errors are at most 0.05.

	$c = 1$					$c = 100$				
k	$n = 0$	$n = 10$	$n = 100$	$n = 400$	$n \to \infty$	$n = 0$	$n = 10$	$n = 100$	$n = 400$	$n \to \infty$
1	0.9	0.8	0.1	0.1	0.1	1.0	1.0	3.5	5.2	6.2
4	3.6	2.9	0.3	0.2	0.2	4.0	4.2	15.5	21.7	25.0
8	7.3	5.1	0.5	0.5	0.5	8.1	9.8	36.2	46.0	50.0
12	11.1	6.7	0.8	0.7	0.7	12.4	23.4	65.1	73.0	74.9
16	15.6	8.0	1.1	0.9	0.9	117	116	109	103	99.9

A.4. The Fully Trusting Predictor

Without demonstrations, a predictor that always treats the description as relevant returns $m^\top x$. The moments used in the proof of Proposition 2 give

$$R^{\mathrm{trust}}(0) = \mathbb{E}\|w - m\|^2 = \big(1 - (2p - 1)(1 - r)\big)a_0 d.$$

For $r < 1$, this risk exceeds the baseline $a_0 d$ exactly when $p < 1/2$. At $r = 1$, the description carries no information, $m = 0$ almost surely, and the two risks coincide for every p.

With demonstrations, the fully trusting predictor uses $\mu_1^\top x$ regardless of Z, where

$$\mu_1 = m + (cI_d + W_n)^{-1} X_n^\top (y_{1:n} - X_n m).$$

For fixed a_0 and n, $\mu_1 \to m$ in mean square as $c \to \infty$. Conditional on relevance, $\mathbb{E}\|w - m\|^2 = r a_0 d \to 0$; conditional on irrelevance, it is $(2 - r)a_0 d \to 2a_0 d$. Thus

$$\lim_{c \to \infty} R^{\mathrm{trust}}(n) = 2(1 - p)a_0 d.$$

For every $p < 1$, this is a positive error floor at fixed n, whereas $R^{\mathrm{demo}}(n) \to 0$ as $n \to \infty$. Hence sufficiently many demonstrations can outperform full trust in a sufficiently precise unreliable description. At $p = 0.9$, $c = 100$, $d = 16$, and $a_0 = 10$, the full-trusting risk is 25.9 against 26.6 without the description at $n = 14$, and 25.5 against 19.5 at $n = 15$; full trust remains worse at every larger n we evaluated, up to 10^5. The order of limits matters: this argument takes $c \to \infty$ at fixed n and does not assert that demonstrations can never overcome a fixed finite prior precision.

B. Additional Transformer Results and Training Details

Evaluation. We evaluate each network on 20,000 fresh prompts. All networks share these prompts, so the standard deviations across seeds exclude prompt sampling variation. At $d = 5$, the demonstration-only learning curve is evaluated at every integer from 0 to 20 (0 to 15 for $c = 0.2$ and $c = 2$). We compute ESS at the first crossing of this curve with the description-conditioned error, as in Definition 1. Table 2 reports results at $n = 0$, including evaluation at reliabilities that differ from the training distribution.

Architecture and optimization. The Transformer has 12 layers, 8 attention heads, and width 256, matching the model size of Garg et al. (2022). We train with Adam at learning rate 10^{-4} for 40,000 steps, using batches of 1,024 newly sampled prompts and no curriculum. Following Zhu et al. (2026), the model predicts a mean and a log variance for the query label and minimizes log loss. The optimal predicted mean is the posterior mean, which is also optimal under squared error; evaluation uses the predicted mean and the loss in Equation (1).

Each demonstration token contains an input and its label. Two indicator coordinates distinguish token types, and the model uses no positional encoding. When a description is absent, its token is masked out of attention. The prediction is read from the query token. Labels and description values are divided by $\sqrt{a_0}$, gradients are clipped to norm 1, and training uses bf16 autocast. The demonstration count is sampled uniformly from 0 to 20, except for the $c = 0.2$ and $c = 2$ experiments below (0 to 15) and $d = 10$ (0 to 40).

Layers are pre-norm, with GELU activations, feed-forward width 1,024, and no dropout; the model has 9.5M parameters at $d = 5$. Adam uses PyTorch's defaults ($\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 10^{-8}$), with no weight decay and a constant learning

Table 2. **Transformer ESS and implied trust under matched and shifted reliability.** Results use $d = 5$, $a_0 = 10$, and no demonstrations, with 20,000 test prompts per network. Rows are grouped by training reliability. "Network" reports mean ESS across the indicated training seeds, with standard deviation; for a single seed, the uncertainty is the standard error over prompts. The train-matched Bayesian predictor uses the training reliability, and the calibrated predictor uses the test reliability. We show a single reference value when they coincide. Seed 0 was not evaluated at $p_{\text{test}} = 0.99$ for training reliabilities 1 and 0.9, so those rows average fewer seeds, and their trust differs from the other rows only through the seeds averaged. Calibrated values, and coincident train-matched values, use 200,000 prompts, with standard errors at most 0.1. Other train-matched values use the network's test prompts; sampling variation can change these estimates by up to 0.6, so an estimated train-matched value can exceed its calibrated counterpart. "Trust" is implied trust averaged across seeds.

				ESS			
c	p_{train}	p_{test}	Seeds	Network	Train-matched	Calibrated	Trust
1	1	1	2	5.85 ± 0.03	5.89	—	0.964
1	1	0.99	1	5.65 ± 0.06	5.66	5.61	0.991
1	1	0.9	2	4.02 ± 0.03	3.98	4.10	0.964
10	1	1	3	14.26 ± 0.48	15.91	—	1.008
10	1	0.99	2	9.38 ± 0.16	9.54	8.96	1.015
10	1	0.9	3	4.58 ± 0.03	4.54	4.70	1.008
1	0.99	1	2	5.87 ± 0.01	5.88	5.89	0.963
1	0.99	0.99	2	5.65 ± 0.01	5.61	—	0.963
1	0.99	0.9	2	4.02 ± 0.03	3.99	4.10	0.963
10	0.99	1	3	13.28 ± 0.57	16.01	15.91	0.982
10	0.99	0.99	3	9.25 ± 0.15	8.96	—	0.982
10	0.99	0.9	3	4.60 ± 0.04	4.56	4.70	0.982
1	0.9	1	2	5.67 ± 0.06	5.75	5.89	0.883
1	0.9	0.99	1	5.51 ± 0.06	5.55	5.61	0.901
1	0.9	0.9	2	4.03 ± 0.01	4.10	—	0.883
10	0.9	1	3	9.49 ± 0.44	10.79	15.91	0.876
10	0.9	0.99	2	8.08 ± 0.28	8.55	8.96	0.880
10	0.9	0.9	3	4.66 ± 0.01	4.70	—	0.876

rate. Training seeds are 0, 1, and 2 for $c = 10$, for $d = 10$, and for the partial description; 0 and 1 for $c = 1$; and 0 for $c \in \{0.2, 2\}$. A $d = 5$ network trains in about 80 minutes and a $d = 10$ network in about 2.8 hours on one NVIDIA L4 GPU; the 25 networks take about 38 GPU-hours.

Without a description, every $d = 5$ network's error is within 0.21 of the Bayes-optimal error of about 50 at $n = 0$, and within 0.32 on average over its training demonstration counts; at $d = 10$ the gaps are at most 0.50 of about 100, and 0.41. With the specific description at $n = 0$, the $c = 10$, $p = 1$ networks' error is 0.67 to 0.71, against 0.49 for the Bayesian predictor: they remove 98.6% of the baseline error, against 99.0%.

Additional specificity levels. Four additional networks cover $c = 0.2$ and $c = 2$, each at $p = 1$ and $p = 0.9$. Their zero-demonstration ESS values are within 0.03 of the Bayesian values on the same prompts: 2.64 versus 2.66, 2.16 versus 2.15, 7.32 versus 7.35, and 4.38 versus 4.37, respectively.

Higher dimension and partial descriptions. At $d = 10$, we train three seeds with the $c = 10$ description and extend training and evaluation to 40 demonstrations so that the baseline curve reaches the ESS crossing. The network ESS is 17.8, compared with a Bayesian value of 20.8 and the limiting expression $c + d + 1 = 21$. As at $d = 5$, the network captures most of the description's value but falls short of the reference.

We also train networks with a partial description, as defined in Appendix A.3. The description token contains m in the first $k = 3$ of the $d = 5$ coordinates and zeros elsewhere; the remaining task coordinates follow the baseline prior. For this $c = 10$ description, mean ESS across three seeds increases from 3.27 at $n = 0$ to 5.55 at $n = 10$. The corresponding Bayesian values are 3.22 and 5.64. This increase agrees with the behavior described by Proposition 4.

ESS with demonstrations. Figure 5 evaluates each network at its training reliability. For $c = 1$, network ESS remains within 0.6 of the Bayesian value at every demonstration count whose crossing lies within the evaluation grid. For $c = 10$, the largest shortfalls are 2.4, 2.9, and 1.4 demonstrations at $p = 1$, 0.99, and 0.9, respectively; the largest excess is 0.3. At $n = 5$, the three networks trained at $p = 0.9$ have ESS values of 4.83, 4.79, and 4.77, compared with the Bayesian value of

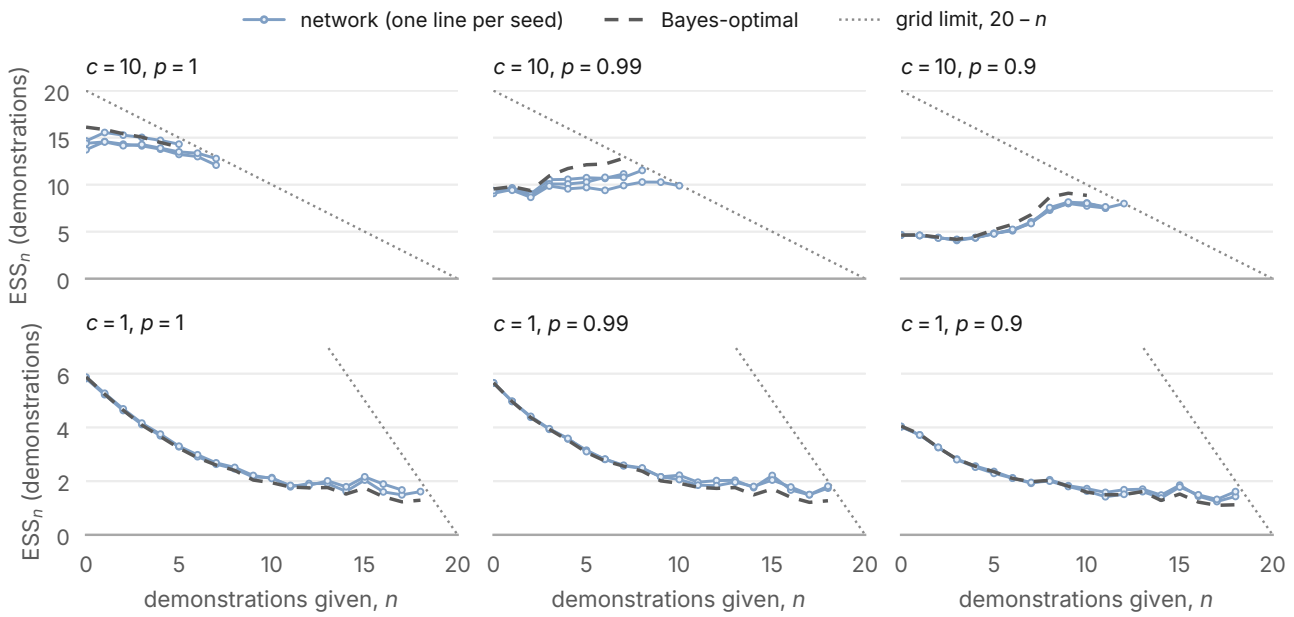


Figure 5. **Transformer ESS as a function of demonstration count.** Each line represents a training seed; the Bayesian reference is evaluated on the same prompts. Training and test reliability are matched. Results cover $c = 10$ and $c = 1$ at $d = 5$, using 20,000 prompts. A curve ends when its crossing exceeds the maximum of 20 demonstrations. The dotted line, $20 - n$, marks the largest ESS identifiable on this grid.

5.20 on the same prompts.

Reliability shift after demonstrations are observed. For $c = 10$, networks trained at $p = 1$ and $p = 0.99$ both behave approximately as if the description were certainly relevant when $n = 0$. Their behavior separates once demonstrations are available. At test reliability $p = 0.9$, the description reduces error for the networks trained at $p = 0.99$ at every evaluated n, as it does for the Bayes-optimal predictor. For networks trained at $p = 1$, the description instead increases error from six demonstrations onward in every seed: the excess error is 0.77 to 1.06 at $n = 6$ and peaks at 2.74 to 2.94 at $n = 9$. A fully trusting Bayesian predictor also first loses at $n = 6$, with excess error 1.12 there and 2.96 at $n = 9$.

C. Additional Language-Model Results and Evaluation Details

We generate 2,000 tasks. Six open-weight models answer all tasks without reasoning; we refer to these experiments as the *full run*. The models are gpt-oss-20b, Qwen3.5 9B, and Gemma 4 31B, which also appear in Section 6, together with Phi-4 (14B; Abdin et al., 2024), OLMo 2 32B Instruct (Team OLMo et al., 2025), and Gemma 4 12B. The experiments in Section 6 use the first 200 tasks. Unless stated otherwise, the results in this appendix use the full run without reasoning.

Table 3 reports summary ESS, defined as the median over $n \in \{1, 2, 4\}$, with uncertainty intervals. Figure 6 shows the underlying learning curves. Tables 12 and 13 report ESS at every evaluated demonstration count; Table 4 compares interpolation methods; Table 14 reports description weights for the models with a distribution readout; and Table 5 evaluates genuinely unreliable descriptions.

Throughout this appendix, the *prompt-based Bayesian reference* uses the probabilistic claims in the prompt together with an explicit assumption for the unspecified fallback prior. It is optimal under that assumed model. When the stated description differs from the data-generating process, it need not minimize error under the evaluation distribution.

C.1. Prompts, Tasks, and Model Readouts

Model inclusion. We retain a model only if it produces a number in at least three quarters of replies overall and, averaged over $n = 1$, 2, and 4 and both noise levels, in at least half of replies without a description, which determine the baseline curve used for summary ESS. This criterion excludes MiniCPM5 2B (MiniCPM Team, 2026), OLMo 2 7B Instruct, and

OLMo 2 13B Instruct, whose replies begin with a number in 62%, 20%, and 10% of cases, respectively. These models often begin with explanations such as "To predict…" or "Given…" despite the numeric-only instruction.

The full run uses Hugging Face Transformers at each model's native precision, with checkpoints `openai/gpt-oss-20b`, `microsoft/phi-4`, `allenai/OLMo-2-0325-32B-Instruct`, `Qwen/Qwen3.5-9B`, `google/gemma-4-12B-it`, and `google/gemma-4-31B-it`. Tasks are sampled by a seeded generator. Reasoning is disabled where the chat template supports this option. For gpt-oss-20b, generation begins in the final channel, after the channel normally used for reasoning.

Distribution and written readouts. For gpt-oss-20b, Phi-4, and OLMo 2 32B without reasoning, the prediction is the mean of the first-token distribution, renormalized over the 900 three-digit number tokens. We call this the *distribution readout*. Other models, and all reasoning runs, use the number in the final answer, called the *written readout*. We also evaluate written readouts for the three distribution-readout models as a robustness check (Table 3 for gpt-oss-20b and Table 6). OLMo 2 32B assigns 41% of first-token probability to numbers without a description at $n = 1$, and 75% at $n = 2$. Its distribution readout at $n = 1$ therefore summarizes a minority of the first-token probability mass.

Table 3. **Description ESS and weights in the full run without reasoning.** Left: median ESS over $n = 1$, 2, and 4, for relevant descriptions with $c = 1$, 4, and 36, and for the $c = 36$ description when irrelevant or accompanied by a stated 10% or 50% chance of being wrong. Brackets give 95% bootstrap intervals. An upper bound, $\leq$, indicates that one demonstration alone outperforms the description-conditioned prediction. Right: the coefficient on the stated value at $n = 1$; the Bayesian reference weight is $c/(c + n)$. All results use $\sigma = 60$. "Written" denotes the written readout.

	ESS, median over $n = 1, 2, 4$						weight, $n = 1$		
	relevant			$c = 36$			relevant		
Model	$c = 1$	$c = 4$	$c = 36$	irrelevant	10% wrong	50% wrong	$c = 1$	$c = 4$	$c = 36$
Reference	1.3	4.4	36	≤ -1.0	22	5.8	0.50	0.80	0.97
gpt-oss-20b	0.9 [0.7, 1.0]	2.0 [1.7, 2.3]	2.8 [2.6, 3.1]	≤ -0.4	2.4 [1.9, 2.6]	2.3 [1.9, 2.7]	0.53	0.48	0.49
Phi-4	0.7 [0.4, 0.9]	4.1 [3.5, 4.6]	7.6 [6.5, 8.6]	≤ -1.0	8.7 [7.8, 9.8]	9.7 [8.7, 11]	0.82	0.76	0.74
OLMo 2 32B	1.9 [1.7, 2.1]	3.4 [3.2, 3.9]	5.3 [4.5, 6.3]	≤ -0.6	6.0 [5.6, 7.1]	5.4 [5.0, 6.0]	0.72	0.70	0.70
gpt-oss-20b (written)	0.2 [0.1, 0.4]	2.0 [1.9, 2.6]	3.7 [3.0, 4.4]	≤ -1.0	2.4 [2.1, 2.6]	2.3 [2.1, 2.6]	0.64	0.52	0.52
Qwen3.5 9B (written)	0.2 [−0.1, 0.4]	2.7 [2.1, 3.2]	4.0 [3.5, 4.5]	≤ -1.0	0.6 [0.3, 1.0]	1.0 [0.6, 1.5]	0.81	0.76	0.80
Gemma 4 12B (written)	0.0 [−0.1, 0.1]	0.5 [0.3, 0.7]	1.8 [1.4, 2.2]	≤ -0.7	1.3 [1.1, 1.6]	2.7 [2.5, 3.1]	0.34	0.24	0.59
Gemma 4 31B (written)	0.0 [0.0, 0.0]	0.2 [0.1, 0.3]	1.4 [1.2, 1.6]	≤ -0.7	0.6 [0.5, 0.7]	0.5 [0.4, 0.6]	0.48	0.49	0.72

Prompt. The following text is supplied as the user turn in the model's chat template:

```
The numbers below are drawn from a normal distribution with standard deviation 60.
Its mean was drawn from a normal distribution with mean 510 and standard deviation 30.
548
431
Predict the next number.  Reply with the number only.
```

Without a description, the second sentence is "Its mean is unknown." To state unreliability, a third sentence is added: "There is a 10% chance that this statement is wrong, in which case the mean is unknown." In the rerun of Section 6 that states the fallback, its last clause becomes "in which case the mean was instead drawn uniformly at random from the whole numbers 100 to 999." With no demonstrations, the final question is "Predict the first number."

Task generation. A relevant description value is sampled conditional on the task so that the joint distribution satisfies $m \sim \mathcal{N}(500, 100^2 - \tau^2)$ and $w \mid m \sim \mathcal{N}(m, \tau^2)$. A task is redrawn if a displayed number or description value falls outside 100 to 999; this occurs for less than 1% of tasks. The full run also includes $\sigma = 30$ conditions. Both standard deviations stated in the prompt are then halved, preserving $c \in \{1, 4, 36\}$. The qualitative pattern is unchanged (Table 12).

Numeric response rates. For the distribution readout, the numeric response rate is the probability mass assigned to number tokens at the first position. For the written readout, it is the fraction of replies containing a three-digit number. Tables 6, 12, 13, and 14 report this quantity as "number". Tables 6 and 14 italicize cells whose rate is below three quarters.

Written-readout implementation and checks. The written readout takes the final answer: the number in the last `\boxed{}` expression if there is one, and otherwise the last number with three digits before an optional decimal part, keeping the decimals. A digit group after a thousands separator, as in 14,945, does not count as a number. In the full run, decoding is greedy with a limit of eight new tokens, so the reply is the number itself; those predictions were recorded with an earlier readout that took the first such number and kept only its integer part. Replies without a qualifying number are excluded from both error curves and weight regressions.

For gpt-oss-20b, written-readout weights differ from distribution-readout weights by at most 0.21 across all conditions and demonstration counts. For $c = 1$, 4, and 36, written-readout summary ESS is 0.2, 2.0, and 3.7, compared with 0.9, 2.0, and 2.8 from the distribution readout. A single greedy response is a noisier summary than the distribution mean. Phi-4 produces a number in only 58% of written replies across all conditions, so Table 3 reports only its distribution readout. In the wording experiment, however, its numeric response rate exceeds three quarters for the original wording. OLMo 2 32B produces a number in only 5%, 36%, and 48% of written replies without a description at $n = 1$, 2, and 4, respectively, with $\sigma = 60$; we therefore do not report its written-readout ESS. Its written-readout weights for the three relevant descriptions at $n = 1$ are 0.83, 0.81, and 0.83, compared with distribution-readout weights of 0.72, 0.70, and 0.70.

C.2. ESS Estimation and the Bayesian Reference

Measurement. We apply Definition 1 to squared error against the hidden mean. In the full run, the baseline curve is evaluated at $n = 1, 2, 4, \ldots, 64$. We interpolate squared error, rather than its square root, linearly in $\log n$. The ESS at n is the first interpolated demonstration count t that matches the description-conditioned error, minus n.

If the description-conditioned error lies below the entire baseline curve up to $t_{\max}$, we report $\mathrm{ESS}_n > t_{\max} - n$. Because the curve starts at one demonstration, we report $\mathrm{ESS}_n \leq 1 - n$ when one demonstration alone already gives lower error. Summary ESS uses $n = 1$, 2, and 4. We exclude $n = 0$ because models rarely return a number without a description or demonstrations. At large n, ESS is strongly affected by slow baseline learning and excessive weight on the $c = 1$ and $c = 4$ descriptions. At $\sigma = 60$ and $n = 64$, model root-mean-square error without a description ranges from 9 to 20, compared with 7.5 for the reference. A learner with a flatter baseline curve can assign a larger ESS to the same error reduction. Description-weight estimates complement ESS because they do not use that curve.

Bootstrap replicates resample tasks jointly for the description-conditioned and baseline curves and recompute the complete ESS statistic. The intervals therefore measure variation across tasks; they do not include interpolation bias, quantified below. We use 500 replicates with a fixed seed, the same resampled tasks for every model.

Reference prior and exact sample equivalence. The prompt-based Bayesian reference takes "unknown" to mean uniform over 100 to 999. Without a description, its prediction is therefore approximately the demonstration mean, with negligible boundary truncation for the evaluated prompts. For a relevant Gaussian description, the idealized continuous model with a flat baseline prior has exact sample equivalence c. Integer rounding, finite-range truncation, and the task-rejection rule are not part of that exact identity; we compute the reported reference values on the same generated tasks as the models. When a chance of being wrong is stated, the reference uses it as a prior probability of irrelevance and updates that probability from the demonstrations. Otherwise, it treats the description as relevant.

Applying the empirical ESS procedure to the full-run reference gives 1.3, 4.4, and 36 for $c = 1$, 4, and 36 at $\sigma = 60$, and 1.2, 4.7, and 43 at $\sigma = 30$. Interpolation in $\log n$ induces an upward bias, while finite-task sampling can shift the estimate in either direction. We use the same estimator for the reference and the models. On the 200 tasks of Section 6, the grid additionally includes $n = 40$ and 48 near the $c = 36$ crossing, and the reference values are 1.0, 3.5, and 34.6. With a stated chance of being wrong, the reference's summary ESS varies more across task samples: on 20,000 tasks it is 7.3 at a stated 50% and 23.4 at 10%, against 5.8 and 22 on the full run's 2,000 tasks and 5.7 and 27.5 on the first 200. Each model is compared with the reference on the same tasks.

The actual task prior, $\mathcal{N}(500, 100^2)$, is not stated in the prompt. A reference using that prior would have baseline precision equivalent to 0.36 demonstrations at $\sigma = 60$ and 0.09 at $\sigma = 30$. At $\sigma = 60$, the corresponding idealized continuous Gaussian model would give a relevant description exact ESS $c - 0.36$, namely 0.64, 3.64, and 35.64. This is the scalar analogue of $c - a_0^{-1}$ in Section 3. We use the prompt-based reference because the models are not given the actual task prior. This distinction matters under misspecification: unlike the correctly specified Bayesian predictor in Sections 3–4, a reference that accepts an incorrect description can have negative ESS, indicating increased prediction error (Reimherr et al., 2021).

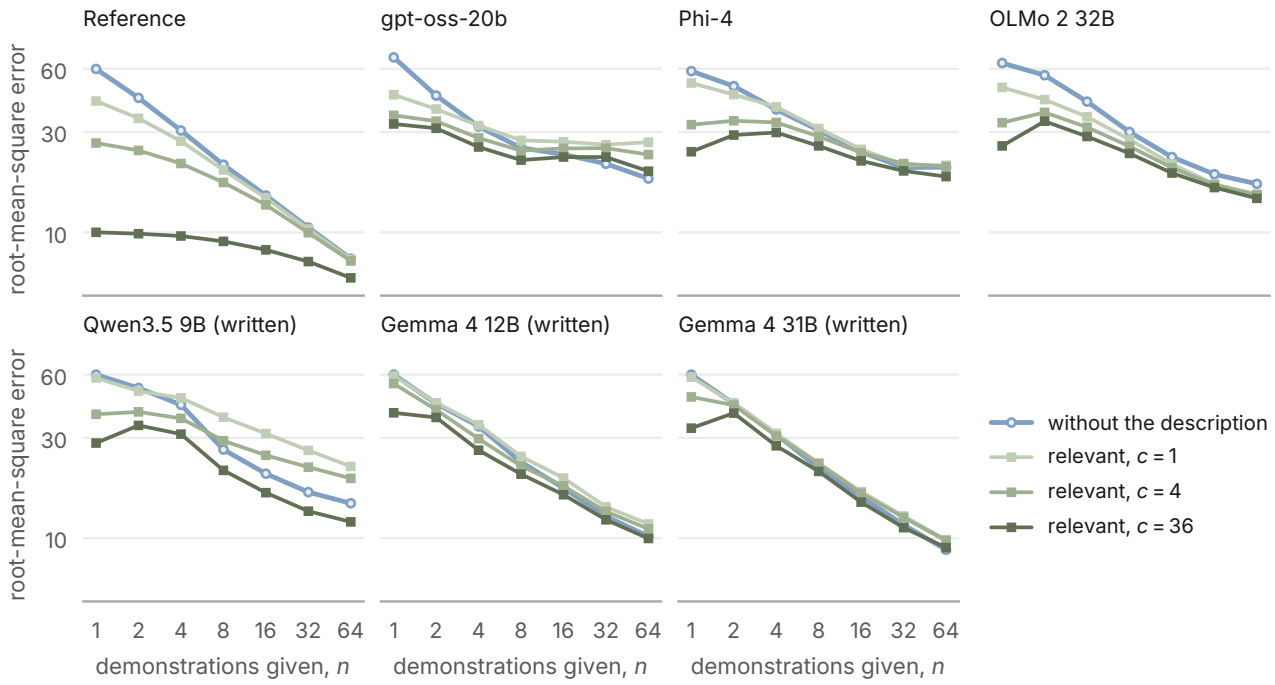


Figure 6. **Learning curves used to estimate language-model ESS.** Root-mean-square error versus demonstration count for the reference and six models in the full run without reasoning, at $\sigma = 60$. Blue curves omit the description; green curves use relevant descriptions, with darker shades indicating higher c. Replies without a number are excluded. ESS is the horizontal distance from the description-conditioned point at n to the baseline curve at the same error. For four models, error with $c = 36$ increases between one and two demonstrations because the models shift toward the observations more than the reference does (Table 14 for the distribution-readout models).

Sensitivity to interpolation. The full-run grid doubles n, so interpolation can affect ESS. On the reference's idealized errors, σ^2/n without a description and $\sigma^2/(c+n)$ with one, interpolation in $\log n$ overestimates ESS by 0.2, 0.3, and 1.8 for $c = 1$, 4, and 36. Linear interpolation in n gives overestimates of 0.3, 0.6, and 4.1.

For the models without reasoning, the relevant crossings are unique. Among the twelve baseline curves across the two noise levels, only Phi-4 at $\sigma = 60$ increases, slightly, from 20.3 at $n = 32$ to 20.4 at $n = 64$. Table 4 compares four estimators: interpolation in $\log n$; linear interpolation in n; a least-squares non-increasing fit using pool adjacent violators; and a relative-least-squares fit of $A/(n + n_0) + B$ to the baseline errors. The final form is a Bayesian learning curve plus an error floor and recovers exact ESS when fit to the reference's idealized errors.

Across these choices, no model's summary ESS changes by more than 1.4 demonstrations. For the $c = 36$ description, no model exceeds 8.1, while the reference ranges from 35 to 38. Differences become larger at higher demonstration counts, where a flat learning curve turns a small error difference into a large change in ESS. For example, gpt-oss-20b at $n = 8$ and $c = 36$ has ESS 17 under the estimator in Table 12, but 9.2 under the fitted curve.

C.3. Unreliability, Description Accuracy, and Wording

Irrelevant descriptions. An irrelevant description uses an independently drawn value. At $c = 36$, these descriptions have negative ESS for every model and for the prompt-based reference, which treats them as relevant. The models assign less weight to the description and consequently incur lower error in this condition: at $n = 4$, their root-mean-square errors range from 35 to 64, compared with 124 for the reference.

Actual unreliability. The stated-unreliability experiments keep descriptions relevant and vary what models are told. Table 5 instead evaluates the mixture in Section 2. We combine squared errors from the relevant and irrelevant conditions, on the same tasks, with weights p and $1 - p$, and then estimate ESS. We evaluate models both with and without disclosure of the irrelevance probability. The fully trusting reference always treats the description as relevant; the calibrated reference starts from irrelevance probability $1 - p$ and updates it from the observations.

Table 4. **Sensitivity of summary ESS to baseline-curve estimation.** Median ESS over $n = 1$, 2, and 4 for relevant descriptions, using the full run without reasoning at $\sigma = 60$. "Log n" is the primary estimator; "linear" interpolates in n; "monotone" applies a non-increasing fit; and "fit" uses $A/(n + n_0) + B$. The reference's idealized ESS values are 1, 4, and 36.

	$c = 1$				$c = 4$				$c = 36$			
Predictor	log n	linear	monotone	fit	log n	linear	monotone	fit	log n	linear	monotone	fit
Reference	1.3	1.4	1.3	1.0	4.4	4.7	4.4	4.0	36	38	36	35
gpt-oss-20b	0.9	1.0	0.9	0.9	2.0	2.4	2.0	2.4	2.8	2.9	2.8	3.4
Phi-4	0.7	0.8	0.7	0.4	4.1	4.4	4.1	3.6	7.6	8.1	7.6	7.1
OLMo 2 32B	1.9	2.1	1.9	1.4	3.4	3.7	3.4	3.3	5.3	5.8	5.3	5.5
Qwen3.5 9B (written)	0.2	0.3	0.2	−0.1	2.7	2.9	2.7	1.3	4.0	4.3	4.0	2.8
Gemma 4 12B (written)	0.0	0.0	0.0	0.0	0.5	0.6	0.5	0.4	1.8	1.9	1.8	1.6
Gemma 4 31B (written)	0.0	0.0	0.0	−0.1	0.2	0.3	0.2	0.0	1.4	1.7	1.4	1.0

Table 5. **Description ESS under actual unreliability.** Relevant and irrelevant conditions, at $c = 4$ and $c = 36$, are combined with weights p and $1 - p$. Results are summary ESS with 95% task-bootstrap intervals for the full run without reasoning at $\sigma = 60$. The fully trusting reference assumes relevance; the calibrated reference uses the actual reliability and updates it from demonstrations. A bound indicates that one demonstration alone outperforms the description-conditioned prediction at one or more summary demonstration counts.

	irrelevant one time in ten ($p = 0.9$)		irrelevant half the time ($p = 0.5$)	
Predictor	$c = 4$	$c = 36$	$c = 4$	$c = 36$
Fully trusting	1.0 [0.8, 1.1]	0.2 [0.0, 0.4]	≤ -1.0	≤ -1.0
Calibrated	2.3 [2.1, 2.7]	5.5 [5.1, 6.0]	0.6 [0.5, 0.7]	0.9 [0.8, 1.1]
gpt-oss-20b, not told	1.2 [1.0, 1.4]	1.6 [1.4, 1.7]	0.1 [0.0, 0.2]	≤ 0.0
gpt-oss-20b, told	1.1 [1.0, 1.3]	1.3 [1.1, 1.4]	−0.1 [−0.1, 0.0]	−0.1 [−0.1, 0.0]
Phi-4, not told	1.3 [1.1, 1.5]	1.9 [1.7, 2.4]	≤ -1.0	≤ -1.0
Phi-4, told	2.7 [2.2, 3.2]	3.6 [3.1, 4.1]	≤ 0.0	≤ 0.0
OLMo 2 32B, not told	2.5 [2.2, 2.8]	3.2 [2.9, 3.5]	≤ 0.7	≤ 0.6
OLMo 2 32B, told	3.2 [2.9, 3.4]	4.0 [3.6, 4.3]	≤ 1.0	≤ 0.9
Qwen3.5 9B (written), not told	1.0 [0.7, 1.2]	1.8 [1.3, 2.1]	≤ -0.3	≤ -0.9
Qwen3.5 9B (written), told	−1.0 [−1.0, −0.7]	−0.1 [−0.5, 0.3]	≤ -1.0	≤ -1.0
Gemma 4 12B (written), not told	0.3 [0.2, 0.5]	0.7 [0.5, 1.0]	≤ -0.1	≤ 0.0
Gemma 4 12B (written), told	0.6 [0.5, 0.7]	1.0 [0.9, 1.3]	≤ 0.0	≤ 0.1
Gemma 4 31B (written), not told	0.1 [0.0, 0.2]	0.4 [0.3, 0.5]	≤ 0.0	≤ -0.1
Gemma 4 31B (written), told	≤ 0.0	0.2 [0.1, 0.3]	≤ 0.0	≤ 0.0

At actual reliability $p = 0.9$, a $c = 36$ description has ESS 5.5 for the calibrated reference and 0.2 for the fully trusting reference. Model ESS ranges from 0.4 to 3.2 when the reliability is not stated and from −0.1 to 4.0 when it is stated. Model weights depend weakly on p, except for Gemma 4 12B at $n = 1$, and stating the probability changes ESS by at most two demonstrations in either direction. At $p = 0.5$, the description is worth less than one demonstration to the calibrated reference and at most one to every model.

Wording without reasoning. We repeat the three relevant-description conditions and the no-description baseline at $n = 1$, 2, and 4 using three alternative description sentences: "Its mean is about m, give or take τ.", "Mean: $m \pm \tau$", and "An earlier estimate put its mean at m, with a standard error of τ." Table 6 uses the full-run tasks and written readouts.

Wording substantially changes the weight assigned to the description. For Gemma 4 12B with $c = 1$ and $n = 1$, this weight is 0.34 under the original wording and 1.00 under each alternative. For gpt-oss-20b, it ranges from 0.17 to 0.67. A coefficient above 1 means that predictions vary more strongly with the stated value than the value itself. Nevertheless, none of the tested wordings yields the reference's sensitivity to specificity: across models and wordings, the weight difference between $c = 36$ and $c = 1$ is at most 0.25 at $n = 1$ and 0.44 at $n = 4$, compared with reference differences of 0.47 and 0.70. The informal descriptions do not uniquely specify a probability distribution, so this comparison also varies how precisely the task is specified.

Separating stated specificity from description accuracy. For each task, we take a description value generated either at $c = 36$ or at $c = 1$ and present it with each stated specificity in $\{1, 4, 36\}$. The tasks and demonstrations match the full

Table 6. **Description weights under four wordings without reasoning.** Weights are reported for the three relevant descriptions at $n = 1$ and 4, using written readouts on the full-run tasks. "Number" denotes the numeric response rate averaged over n; a weight is italicized when the numeric response rate at its n is below three quarters. The reference weights are 0.50, 0.80, and 0.97 at $n = 1$, and 0.20, 0.50, and 0.90 at $n = 4$. Wording changes overall reliance on the description, but the observed change with stated specificity remains smaller than the reference change.

			weight, $n = 1$			weight, $n = 4$		
Model	Wording of the description	number	$c = 1$	$c = 4$	$c = 36$	$c = 1$	$c = 4$	$c = 36$
Gemma 4 12B	normal, mean m, sd τ (ours)	1.00	0.34	0.24	0.59	0.19	0.24	0.31
	about m, give or take τ	1.00	1.00	1.00	1.00	0.87	0.76	0.82
	Mean: $m \pm \tau$	1.00	1.00	1.00	1.00	0.60	0.66	0.74
	earlier estimate m, s.e. τ	1.00	1.00	1.00	1.00	0.45	0.54	0.67
Gemma 4 31B	normal, mean m, sd τ (ours)	1.00	0.48	0.49	0.72	0.02	0.04	0.18
	about m, give or take τ	1.00	1.00	1.00	1.00	1.01	1.22	1.45
	Mean: $m \pm \tau$	1.00	1.00	1.00	1.00	0.90	1.03	1.15
	earlier estimate m, s.e. τ	1.00	0.94	0.94	0.96	0.09	0.23	0.41
gpt-oss-20b	normal, mean m, sd τ (ours)	1.00	0.64	0.52	0.52	0.23	0.29	0.34
	about m, give or take τ	1.00	0.24	0.17	0.38	0.28	0.29	0.38
	Mean: $m \pm \tau$	1.00	0.17	0.26	0.40	0.23	0.24	0.30
	earlier estimate m, s.e. τ	1.00	0.67	0.63	0.67	0.28	0.37	0.42
OLMo 2 32B	normal, mean m, sd τ (ours)	0.61	0.83	0.81	0.83	*0.20*	*0.29*	0.38
	about m, give or take τ	0.82	0.93	0.92	0.96	0.60	0.57	0.65
	Mean: $m \pm \tau$	0.82	0.88	0.83	0.88	0.44	0.46	0.55
	earlier estimate m, s.e. τ	0.82	0.95	0.94	0.95	0.49	0.57	0.65
Phi-4	normal, mean m, sd τ (ours)	0.77	0.96	0.78	0.76	*0.38*	*0.37*	*0.38*
	about m, give or take τ	0.97	1.02	1.01	1.02	0.65	0.57	0.62
	Mean: $m \pm \tau$	0.83	*0.99*	*0.87*	*0.87*	0.45	0.42	0.50
	earlier estimate m, s.e. τ	0.81	0.88	0.76	0.73	*0.47*	*0.49*	0.54
Qwen3.5 9B	normal, mean m, sd τ (ours)	1.00	0.81	0.76	0.80	0.26	0.27	0.44
	about m, give or take τ	1.00	0.99	1.00	0.99	0.35	0.43	0.57
	Mean: $m \pm \tau$	1.00	1.08	1.25	1.14	0.19	0.27	0.38
	earlier estimate m, s.e. τ	1.00	0.87	0.86	0.86	0.61	0.75	0.82

run, and the experiment includes its own no-description prompts at every n (Table 7). Repeated conditions reproduce the full run exactly except for gpt-oss-20b. Its distribution readout changes by a median of 0.8 to 1.2 across conditions, with a maximum difference of 39; its written replies change in 12% of cases, with median difference 0. A newer library version is a possible explanation.

The prompt-based reference accepts the stated specificity. A value generated at $c = 1$ but stated as $c = 36$ therefore has reference ESS of about -0.8; a model that discounts the stated specificity can outperform it in that misspecified condition. For models without reasoning, description accuracy primarily determines ESS. When the value is generated at $c = 1$, each model's ESS remains between -0.1 and 2.0 across all stated specificities.

C.4. Reasoning Runs and Execution Settings

Shared evaluation prompts. Every model in Figure 4, as well as Olmo 3 7B Think (Team Olmo, 2025), is evaluated on the same 7,800 prompts from the first 200 tasks. These include no-description prompts at $n = 1$, 2, 4, ..., 64, with additional points at $n = 40$ and 48 near the $c = 36$ crossing, and every description condition at $n = 1$, 2, and 4. The gpt-oss-20b and Olmo 3 runs omit the two additional points and contain 7,400 prompts. Results without reasoning for the open-weight models are taken from the full run restricted to these tasks. The two modes receive the same prompts, although their serving environments can differ as detailed below. At least 99.7% of replies in each run contain a number.

Open-weight reasoning models. With reasoning, gpt-oss-20b runs in vLLM using greedy decoding, low reasoning effort, and at most 4,096 new tokens. It reaches the final channel in 99.9% of replies and uses 322 tokens on average. Qwen3.5 9B runs through OpenRouter at temperature 0, with reasoning enabled and a limit of 24,576 new tokens. We pin requests to DeepInfra, which serves the model in bf16, because quantization varies across providers. Of its replies, 2,305 loop to the

Table 7. **Sensitivity to stated specificity with the value and demonstrations held fixed.** Results use $\sigma = 60$, the full-run tasks, and the readout conventions of Table 3. For a value generated at $c = 36$, the table reports its weight at $n = 1$ and summary ESS under each stated c. Results for the 200-task runs of Section 6 appear in Table 8.

	weight, $n = 1$			ESS		
stated c	1	4	36	1	4	36
Reference	0.50	0.80	0.97	2.5	13	36
gpt-oss-20b	0.48	0.47	0.49	2.7	2.6	2.8
Phi-4	0.81	0.75	0.74	12	7.2	7.6
OLMo 2 32B	0.70	0.70	0.70	4.8	4.6	5.3
gpt-oss-20b (written)	0.52	0.46	0.50	2.9	3.5	3.8
Qwen3.5 9B (written)	0.76	0.75	0.80	3.3	3.5	4.0
Gemma 4 12B (written)	0.38	0.29	0.59	1.0	1.0	1.8
Gemma 4 31B (written)	0.43	0.49	0.72	0.2	0.3	1.4

token limit. Each is rerun once with the model card's recommended reasoning sampling settings (temperature 1.0, top-p 0.95, top-k 20, presence penalty 1.5); 11 remain unfinished and one request fails.

Gemma 4 31B (`gemma-4-31b-it`) runs through the Gemini API at temperature 0. Reasoning is enabled by default, and the model uses 2,613 reasoning tokens on average, with the same 24,576-token limit. Of its replies, 435 reach the limit; all finish after one rerun with the model card's recommended sampling settings (temperature 1.0, top-p 0.95, top-k 64).

Olmo 3 7B Think (`allenai/Olmo-3-7B-Think`) runs in vLLM with greedy decoding and at most 32,768 new tokens. We rerun the 169 replies that reach this limit once with temperature 0.6 and top-p 0.95; 17 remain unfinished. The model generates 9,215 tokens on average and writes a worked solution before most answers. Its long greedy outputs are sensitive to hardware: an earlier stated-specificity run on a different GPU produces the same final number on only half the prompts. We therefore report one run. It averages the demonstrations about as well as the reference but relies on the description more at $c = 1$ and 4, and at $c = 36$ its replies split between the description value alone and other rules; its prediction behavior is reported below.

Models served through command-line agents. GPT-6 Sol (`gpt-6-sol`, the version Codex served during the runs; OpenAI, 2026) runs through the Codex command-line agent, in a read-only sandbox, with an empty configuration directory, so that no user instructions or settings load, and with every tool that can run code disabled. We evaluate reasoning disabled, with no reasoning tokens, and reasoning enabled at low effort. An earlier configuration loaded user instructions and allowed shell commands, which the model used in about 8% of replies, mostly to average long demonstration sequences. Without reasoning, that configuration also gave the description much less weight (ESS 1.5 rather than 10.3 at $c = 36$); with reasoning, every summary ESS changes by at most 2.7. The earlier configuration also changed how often GPT-6 Sol reasons at low effort: without a description it reasoned before 51%, 63%, and 89% of replies at $n = 1$, 2, and 4, against 0.5%, 4%, and 13% in the final runs, and before every reply with a stated 50% chance at $n = 1$, against 74%. The wording runs below used the earlier configuration, and the model invoked a shell in 4 of their 5,400 replies.

Claude Opus 4.6 (`claude-opus-4-6`; Anthropic, 2026b) and Haiku 5.5 (`claude-haiku-5-5`; Anthropic, 2026a) run through the Claude Code command line, with one prompt per call, no tools, no project or user instructions, and the system prompt "You are a helpful assistant." Opus 4.6 uses high effort in both evaluated modes, with reasoning either enabled or disabled; the disabled mode emits no reasoning tokens. When enabled, it reasons before at least 99% of replies in each description condition and averages 599 reasoning tokens across all replies. Without a description, it reasons before only 1%, 24%, and 79% of replies at $n = 1$, 2, and 4, respectively. Haiku 5.5 is evaluated at medium and low effort. Both Claude models use the command line's default sampling settings, which it does not report, and one reply per prompt. Without reasoning, Opus 4.6 still writes visible working in 64 of its 7,800 replies; this changes no summary ESS by more than 0.05.

Run dates. The full run took place on 2–3 October 2026 and its control conditions and gpt-oss-20b's reasoning run on 20 October; Haiku 5.5 on 20 October (main runs) and 4 November (wording, stated fallback, and the semantic task); Opus 4.6, Qwen3.5 9B, Gemma 4 31B, and Olmo 3 7B Think on 3–4 November; and GPT-6 Sol on 3 November (wording) and 4 November (main runs, stated fallback, and the semantic task). The GPT-6 Sol logs do not record a model version.

Table 8. **Description ESS across reasoning configurations.** Results use the first 200 tasks at $\sigma = 60$. Summary ESS is computed using each run's baseline curve; parentheses show the value using the Bayesian reference's baseline where it differs. "10% wrong" and "50% wrong" denote the stated probability for a $c = 36$ description. The last column uses a value generated at $c = 36$ but described as $c = 4$. Each run names the mode and, in parentheses, the provider's effort setting. An upper bound, $\leq$, indicates that one demonstration alone gives lower error; > 60 denotes a crossing beyond the grid. Reference values use the same tasks and estimator. Values without reasoning for gpt-oss-20b, Qwen3.5 9B, and Gemma 4 31B are the open points of Figure 4, and on all 2,000 tasks in Table 3. The runs also include three further fixed-value conditions, not shown.

Run	$c = 1$	$c = 4$	$c = 36$	Irrelevant	10% wrong	50% wrong	Stated $c = 4$ True $c = 36$
Reference	1.0	3.5	34.6	≤ -1.0	27.5	5.7	13.7
GPT-6 Sol, without reasoning	0.9	1.9	10	≤ -1.0	3.3	0.9	2.4
GPT-6 Sol, with reasoning (low)	0.9	3.3	35 (34)	≤ -1.0	30 (29)	4.2	11
Opus 4.6, without reasoning	0.9	2.0	5.8 (5.9)	≤ -1.0	6.1 (6.0)	3.8	3.0
Opus 4.6, with reasoning (high)	1.0	3.5	35	≤ -1.0	29	3.8	13
Gemma 4 31B, with reasoning	1.0	3.5	35	≤ -1.0	21	4.0	14
Qwen3.5 9B, with reasoning	0.9	3.5	> 60 (34)	≤ -1.0	29 (27)	3.9	15 (14)
gpt-oss-20b, with reasoning (low)	0.3	1.0	> 60 (24)	≤ -1.0	1.3 (1.2)	8.8	8.6 (8.4)
Haiku 5.5, with reasoning (medium)	1.1	3.3 (3.5)	28	≤ -1.0	33	3.8 (3.9)	8.5 (8.8)
Haiku 5.5, with reasoning (low)	0.8 (0.7)	3.2 (3.3)	24	≤ -1.0	27 (26)	5.5 (5.6)	8.6 (8.4)
Olmo 3 7B Think, with reasoning	-0.5	0.0	6.2 (6.0)	≤ -1.0	5.9 (5.6)	5.5 (5.3)	15 (13)

Interpreting the comparison between modes. The experiments hold the model identifier and prompts fixed within each paired comparison, but reasoning settings, serving software, readout, retry policy, and access to calculation tools are not uniform across all runs. The results describe the behavior of the evaluated configurations. They do not by themselves isolate a single internal mechanism responsible for the difference between modes.

ESS when baseline arithmetic fails. With reasoning, gpt-oss-20b and Qwen3.5 9B assign approximately the reference weight to the description: at $n = 4$ and $c = 36$, their weights are 0.89 and 0.90, compared with 0.90 for the reference. However, their errors without a description stop falling at larger n. gpt-oss-20b makes errors when averaging long demonstration sequences: its no-description root-mean-square error is 19.0 at $n = 32$ and 24.7 at $n = 64$, compared with reference errors 10.0 and 7.4, and 37 and 62 of its replies at these counts differ from the sample mean by more than 20. Qwen3.5 9B levels off near 11 from $n = 32$ onward (11.2 at $n = 64$), with up to 11 replies per cell that far from the sample mean.

For these models, some description-conditioned errors never meet the baseline curve within the evaluated grid. Table 8 therefore also reports, in parentheses, a comparison using the Bayesian reference's baseline curve. This quantity is the number of demonstrations required by the reference without a description to match the model's error, rather than ESS in the model's own units. The gpt-oss-20b model also makes arithmetic errors with descriptions, reducing the reported ESS in those conditions; for example, it obtains 1.3 when the prompt states a 10% chance of being wrong.

Opus 4.6 and GPT-6 Sol do not exhibit these arithmetic failures. Opus 4.6 has root-mean-square error 7.4 at $n = 64$, matching the reference on these tasks, and using either baseline changes their ESS by at most 1. At low effort, Haiku 5.5 often skips reasoning and repeats the demonstration. Its description weight at $n = 1$ is consequently low even in conditions whose ESS is close to the reference value.

C.5. Prediction Rules and Reliability Updating

Rules inferred from individual replies. At $n = 4$, we estimate the description weight implied by an individual prediction as $(\hat{y} - \bar{y})/(m - \bar{y})$. We use the 61 tasks per cell where $|m - \bar{y}| \geq 30$ and exclude replies without a number. We classify a reply as consistent with a rule if its implied weight is within 0.05 of that rule's weight. The candidate weights are the Bayesian reference (0.90 for $c = 36$ and 0.20 for $c = 1$); the reference with its two precisions exchanged (0.10 and 0.80); the description counted as one additional observation, $1/(n + 1)$; the description alone, with weight 1; and the demonstrations alone, with weight 0. The one-additional-observation rule coincides with the reference at $c = 1$.

With reasoning, Opus 4.6, GPT-6 Sol, Gemma 4 31B, and Haiku 5.5 match the reference rule on 89 to 100% of eligible

replies. Without reasoning, 77% of Opus 4.6 replies at $c = 36$ match none of these candidate rules, as do more than half of GPT-6 Sol replies. Olmo 3 7B Think instead repeats the description value in 57 to 80% of replies at $n = 1$, more often the higher the stated specificity. At $n = 4$, about half of its eligible replies return the description value alone and a third match the reference. Without a description it averages the demonstrations well: its root-mean-square error falls from 56 at $n = 1$ to 10.8 at $n = 32$, against 10.0 for the reference. Its shortfall therefore lies in combining the description with the demonstrations, and reasoning mode is not sufficient for the desired behavior in every tested model.

A fixed reliability rule. Table 9 reports description weights for $c = 36$ when the prompt states a probability of being wrong. A predictor that combines the description-conditioned and no-description predictions with fixed weights p and $1 - p$ assigns weight $p\,36/(36 + n)$ to m. Under the original warning, which leaves the fallback unstated, GPT-6 Sol, Gemma 4 31B, and Qwen3.5 9B with reasoning are close to this fixed rule. Opus 4.6 and Haiku 5.5 at medium effort fall below the reference weight by $n = 4$. Haiku 5.5 at low effort, which often skips reasoning at $n = 1$, comes closest to the reference weight. These are behavioral comparisons; they do not identify the internal procedure used in every reply.

Sensitivity to the unspecified fallback prior. The prompt states only that, if the description is wrong, "the mean is unknown". A Bayesian relevance update compares the likelihood of the observations under the description with their likelihood under this fallback. Because the prompt leaves the fallback distribution unspecified, it does not determine a unique Bayesian relevance update.

For the $c = 36$ description with a stated 50% chance of being wrong, the reference's fitted weights on the first 200 tasks at $n = 1$, 2, 4, 8, and 16 are 0.58, 0.61, 0.64, 0.58, and 0.54 with the uniform fallback over 100 to 999. They are 0.49, 0.49, 0.53, 0.49, and 0.45 with the actual task prior $\mathcal{N}(500, 100^2)$; 0.52, 0.54, 0.58, 0.53, and 0.49 with $\mathcal{N}(500, 200^2)$; and 0.72, 0.74, 0.74, 0.67, and 0.60 with a uniform fallback over 0 to 2,000. The fixed rule gives 0.49, 0.47, 0.45, 0.41, and 0.35.

By four demonstrations, the reference weight exceeds the fixed-rule weight for every fallback examined. Thus the observed agreement of several models with the fixed rule persists across these choices of reference. The size of their departure from Bayesian updating depends on the fallback, however; under the actual task prior, the reference update is small before four demonstrations.

Table 9. **Description weights under stated unreliability.** Coefficients on the $c = 36$ description value for the first 200 tasks, with stated irrelevance probability $1 - p$. "Fixed" uses weight $p\,36/(36 + n)$; the reference updates relevance using the uniform fallback over 100 to 999. Bootstrap standard errors are at most 0.03 at $n = 4$ in the stated-50% condition, and at most 0.06 overall, except for gpt-oss-20b, whose maximum is 0.09. At the stated 50%, several reasoning configurations closely track the fixed rule.

	50% wrong			10% wrong		
	$n = 1$	2	4	1	2	4
Reference	0.58	0.61	0.64	0.88	0.87	0.84
Fixed	0.49	0.47	0.45	0.88	0.85	0.81
GPT-6 Sol	0.49	0.46	0.40	0.90	0.86	0.71
Opus 4.6	0.56	0.43	0.40	0.93	0.86	0.77
Gemma 4 31B	0.49	0.46	0.45	0.89	0.68	0.76
Qwen3.5 9B	0.49	0.46	0.43	0.88	0.84	0.80
Haiku 5.5, medium	0.55	0.56	0.46	0.79	0.93	0.89
Haiku 5.5, low	0.49	0.62	0.58	0.42	0.94	0.88
gpt-oss-20b	0.89	0.87	0.82	0.95	0.87	0.90

Stating the fallback. To remove this ambiguity, we rerun the stated-reliability condition on the same 200 tasks, for $c = 36$, stated chances of 10% and 50%, and $n = 1$, 2, 4, with the fallback stated as the reference's uniform over 100 to 999. We use GPT-6 Sol at low effort and Haiku 5.5 at medium effort, with 1,200 prompts each. Table 10 compares their description weights with those of the original runs. With a stated 50% chance, the reference and the fixed rule differ by at least 5 on 123 to 155 tasks per cell. On those tasks, the share of replies closer to the reference than to the fixed rule rises from 0–5% to 92–97% for GPT-6 Sol and from 21–41% to 87–95% for Haiku 5.5. With a stated 10% chance, at most 7 tasks per cell separate the two rules, so this share is uninformative there.

A rule that trusts the description by a constant amount, without updating it, could also pass this comparison, because on most of these tasks the reference's trust is well above the stated 50%. A sharper test uses each reply's implied trust: its implied

weight on m divided by $c/(c+n)$, on tasks where $|m-\bar{y}| \geq 30$. The reference's trust varies across these tasks from 0.08 to 0.88, according to how well the demonstrations agree with the description, whereas a rule that does not update keeps it constant. Regressing the replies' implied trust on the reference's, pooled over $n = 1$, 2, and 4 at the stated 50% (269 replies per model), gives slopes of 0.96 $[0.84, 1.04]$ for GPT-6 Sol and 0.98 $[0.82, 1.15]$ for Haiku 5.5 with the fallback stated, and 0.01 $[-0.09, 0.11]$ and 0.22 $[-0.01, 0.45]$ under the original warning (95% intervals from resampling the 153 tasks). With the fallback stated, the intercepts are 0.01 and 0.03 and R^2 is 0.57 and 0.33. The test does not rest on the bulk of tasks where trust is high: on the 23 replies where the reference's trust is below 0.5 (mean 0.33), the models' mean implied trust is 0.30 and 0.38 with the fallback stated, against 0.48 and 0.50 under the original warning. With the fallback stated, both models therefore update relevance task by task, about as much as the reference does.

Summary ESS, computed against each model's curve without the description from the main run, moves at 50% from near the fixed rule's value, 4.4, toward the reference's 5.7: from 4.2 to 5.4 for GPT-6 Sol and from 3.8 to 4.2 for Haiku 5.5. At 10%, it changes from 29.7 to 25.0 for GPT-6 Sol and from 32.6 to 15.1 for Haiku 5.5, against 27.5. The two models' weights agree within 0.01 (Table 10), so Haiku's lower ESS does not reflect lower average reliance; it comes from a minority of replies that stray far from the reference, some of which repeat the demonstration.

Table 10. **Stating the fallback changes the reliability rule.** Coefficients on the $c = 36$ description value for the first 200 tasks, as in Table 9. "Unstated": the original warning, "the mean is unknown"; "stated": the mean is otherwise uniform over 100 to 999, as the reference assumes.

	50% wrong			10% wrong		
	$n=1$	2	4	1	2	4
Reference	0.58	0.61	0.64	0.88	0.87	0.84
Fixed	0.49	0.47	0.45	0.88	0.85	0.81
GPT-6 Sol, unstated	0.49	0.46	0.40	0.90	0.86	0.71
GPT-6 Sol, stated	0.58	0.59	0.59	0.88	0.83	0.80
Haiku 5.5, unstated	0.55	0.56	0.46	0.79	0.93	0.89
Haiku 5.5, stated	0.62	0.60	0.63	0.89	0.84	0.80

C.6. Wording with Reasoning Enabled

GPT-6 Sol. We evaluate the three alternative wordings at low effort for the three relevant descriptions and $n = 1$, 2, and 4, using the first 200 tasks. Of $5{,}400$ replies, $5{,}399$ contain a number. Under "An earlier estimate put its mean at m, with a standard error of τ", the model reasons before every reply, and its description weight is within 0.01 of the reference in every cell.

Under "Its mean is about m, give or take τ" and "Mean: $m \pm \tau$", it skips reasoning in 24 to 51% and 11 to 50% of replies per cell, respectively. Of those replies, 83% and 87% repeat the description value. At $c = 1$ and $n = 1$, the fitted description weights are consequently 0.89 and 1.00, compared with reference weight 0.50. Among replies that do include reasoning, 68% and 43% are within 1 of the reference prediction, while 36% and 68% repeat the description value. These categories overlap at $c = 36$, where the description value and reference prediction can be close.

Haiku 5.5. We evaluate the same alternatives at medium effort; all $5{,}400$ replies contain a number. Under "An earlier estimate put its mean at m, with a standard error of τ", the model reasons before at least 71% of replies in each cell, and 99% of the replies with reasoning lie within 1 of the reference prediction. Under "Its mean is about m, give or take τ" and "Mean: $m \pm \tau$", it skips reasoning in 4 to 36% and 2 to 74% of replies per cell, respectively. Even among replies with reasoning, 71% and 91% repeat the description value, while 31% and 20% lie within 1 of the reference prediction. At $c = 1$ and $n = 4$, its description weights are 0.72 and 0.93, compared with reference weight 0.20.

These wording changes affect the precision of the probabilistic specification as well as the model's behavior. All provide a value m and a scale τ, but "give or take" and "$\pm$" do not uniquely identify τ as a prior standard deviation or establish a sampling model. The experiment therefore shows that accurate combination of descriptions and demonstrations under explicit statistical statements does not automatically transfer to these less explicit formulations. It does not isolate sensitivity to semantically equivalent descriptions.

C.7. A Semantic Task

Task. A message has six binary properties: the sender is outside the organization; it requests a payment; it concerns travel; it requests computer access; it has a deadline within a day; and it contains personal contact information. Each message states every property one way or the other, in one of three phrasings per value, such as "I'm a contractor from another company." or "I'm on the sales team here." Each property is present independently with probability one half, the phrasing is drawn uniformly, and the sentences appear in random order. A policy uses exactly two properties and requires each to be present or absent; a message goes into folder A when both requirements hold, and into B otherwise. The prompt describes this family of policies and states that the company's policy was chosen at random, with each of the 60 possible policies equally likely. It then gives the policy note, n previously filed messages with their folders, and four new messages, and asks for the probability that each new message goes into A, as a number between 0 and 1, in four lines from "1: ¡probability¿" to "4: ¡probability¿". Without reasoning, GPT-6 Sol reads each property correctly in 97 to 100% of 96 messages; all three misreadings concern one phrasing, "I'm not traveling for this."

Policy notes. The broad note reads, for example, "Policy note: whether it concerns travel is one of the two properties the policy uses," which leaves 20 policies. The specific note reads "Policy note: to go into A, a message must request a payment. The other property the policy uses is whether it concerns travel," which leaves 2. The unreliable note adds "This note has a 50% chance of being wrong, in which case it says nothing about the policy, which is then any of the 60, equally likely." It is wrong with probability one half (in 92 of the 200 tasks), when it describes an independently drawn policy.

Reference and ESS. The reference's probability of A is the share of policies consistent with the note and the filed messages that send the new message to A. For the unreliable note, it mixes these policies with those consistent with the filed messages alone, weighting the note by the stated 50% updated on the demonstrations. On 3,000 simulated tasks, it is calibrated in every condition, and its posterior probability that the unreliable note is right matches how often it is. The loss is the squared error of the probability against the true folder, averaged over the four messages. ESS follows Definition 1, against each learner's own curve without a note at $n = 0$, 2, 4, 6, 8, 12, 16, 20, 24, and 32, interpolated linearly; on simulated tasks, this grid moves the reference's ESS by at most 0.6 from its value read at every n.

Runs. We use 200 tasks, each a policy draw, and 19 cells: no note at the ten demonstration counts above, and each note at $n = 0$, 2, and 4; this gives 3,800 prompts per configuration. Haiku 5.5 runs at medium effort and GPT-6 Sol without reasoning and at low effort, through the command-line agents of Appendix C.4, with no tools. No reply called a tool. Of the replies, 14 from GPT-6 Sol at low effort give no probability and are left out, and one from Haiku 5.5 omits one of its four. Haiku 5.5 reasons before every reply, with a median of 3,008 reasoning tokens; GPT-6 Sol at low effort reasons before 91%, with a median of 142.

Results. Table 11 reports the ESS of each note. With reasoning, both models are within 2 of the reference in every cell but one, GPT-6 Sol's specific note at $n = 4$: 26.1 against 14.8. Its loss with that note, 0.008, is close to the reference's 0.007, but its own curve without a note flattens past $n = 16$, with loss 0.022 at $n = 20$ against 0.004, so it needs more of its own demonstrations to match the note. For the unreliable note, we compare predictions with the reference and with a fixed rule that keeps the stated 50% unless the filed messages rule out the note, on the messages where the two differ by at least 0.05 (about 150 at $n = 2$ and 200 at $n = 4$, of 800). The share of predictions closer to the reference is 90% at $n = 2$ and 99% at $n = 4$ for Haiku 5.5, 77% and 87% for GPT-6 Sol at low effort, and 45% and 45% for GPT-6 Sol without reasoning. Without reasoning, GPT-6 Sol learns slowly from demonstrations: its loss without a note is still 0.078 at $n = 32$, against 0.000 for the reference. In its own unit, a note can therefore be worth as many demonstrations as to the reference or more, such as 13.5 for the specific note at $n = 2$, although its loss with that note is 0.130, against 0.027.

C.8. Complete Tables

Tables 12–14 report ESS and description weights at every demonstration count for the full run without reasoning.

Table 11. **ESS of each policy note in the semantic task.** ESS on 200 tasks, against each learner's own curve without a note. "≤ 0": the note did not lower the loss; "> 28": the curve without a note does not reach the loss with the note by $n = 32$. On 6,000 simulated tasks, the reference's ESS is 9.6 and 13.2 for the specific note at $n = 2$ and 4, and 2.5 and 2.9 for the unreliable note; models are compared with the reference on the same 200 tasks. Paired 95% bootstrap intervals over tasks (1,000 replicates) for the specific note at $n = 2$: reference $[9.7, 12.9]$, Haiku 5.5 $[8.7, 11.5]$, GPT-6 Sol at low effort $[7.8, 12.7]$, and without reasoning $[3.7, 21.9]$.

Note	n	Reference	Haiku 5.5, medium	GPT-6 Sol, low	GPT-6 Sol, no reasoning
Broad	0	0.0	0.0	≤ 0	≤ 0
	2	1.4	1.3	1.3	0.7
	4	1.8	1.6	1.7	8.6
Specific	0	3.7	3.6	3.3	3.7
	2	11.4	9.9	9.5	13.5
	4	14.8	14.5	26.1	> 28
Unreliable	0	1.3	1.3	1.6	1.7
	2	3.0	2.6	3.2	1.7
	4	5.0	4.9	5.1	2.7

Table 12. **ESS by description condition and demonstration count: distribution readout.** Results cover the prompt-based reference and the three distribution-readout models in the full run without reasoning. "Number" is the numeric response rate averaged over n. "10% wrong" and "50% wrong" denote stated irrelevance probabilities. An upper bound, $\leq$, marks error above the first baseline point; $> 64 - n$ means that the description-conditioned error lies below the entire baseline curve through 64 demonstrations.

Predictor	Description	number	$n = 1$	$n = 2$	$n = 4$	$n = 8$	$n = 16$	$n = 32$	$n = 64$
Reference	$\sigma = 60, c = 1$, relevant		1.2	1.3	1.3	1.3	1.2	1.5	> 0
	$\sigma = 60, c = 1$, irrelevant		≤ 0.0	-0.5	-0.8	-1.2	-1.7	-1.5	-1.7
	$\sigma = 60, c = 4$, relevant		4.5	4.4	3.9	4.6	4.6	5.2	> 0
	$\sigma = 60, c = 4$, irrelevant		≤ 0.0	≤ -1.0	≤ -3.0	-6.2	-12	-20	-32
	$\sigma = 60, c = 36$, relevant		36	36	37	38	39	> 32	> 0
	$\sigma = 60, c = 36$, irrelevant		≤ 0.0	≤ -1.0	≤ -3.0	≤ -7.0	≤ -15	≤ -31	-62
	$\sigma = 30, c = 1$, relevant		1.0	1.2	1.4	1.6	1.4	1.5	> 0
	$\sigma = 30, c = 4$, relevant		4.5	4.7	5.0	5.1	4.9	5.6	> 0
	$\sigma = 30, c = 36$, relevant		42	43	43	43	42	> 32	> 0
	$\sigma = 60, c = 4$, relevant, 10% wrong		3.8	3.9	3.6	4.4	4.5	5.1	> 0
	$\sigma = 60, c = 4$, irrelevant, 10% wrong		≤ 0.0	-0.7	-1.2	-1.6	-2.2	-1.9	-2.2
	$\sigma = 60, c = 36$, relevant, 10% wrong		13	22	26	35	38	> 32	> 0
	$\sigma = 60, c = 36$, irrelevant, 10% wrong		≤ 0.0	-0.9	-1.7	-2.8	-5.4	-6.9	-10
	$\sigma = 60, c = 4$, relevant, 50% wrong		1.9	2.2	2.9	3.7	4.1	4.5	> 0
	$\sigma = 60, c = 4$, irrelevant, 50% wrong		≤ 0.0	-0.2	-0.4	-0.5	-0.8	-0.6	-0.7
	$\sigma = 60, c = 36$, relevant, 50% wrong		3.1	5.8	12	21	31	> 32	> 0
	$\sigma = 60, c = 36$, irrelevant, 50% wrong		≤ 0.0	-0.3	-0.7	-1.2	-2.4	-2.8	-4.5
gpt-oss-20b	$\sigma = 60, c = 1$, relevant	0.98	1.0	0.9	-0.1	-1.5	-9.2	-25	-57
	$\sigma = 60, c = 1$, irrelevant	0.98	≤ 0.0	-0.3	-1.5	-3.6	-10	-25	-59
	$\sigma = 60, c = 4$, relevant	0.98	2.3	1.7	2.0	2.9	-7.5	-24	-48
	$\sigma = 60, c = 4$, irrelevant	0.97	≤ 0.0	-0.3	-1.4	-3.7	-11	-26	-59
	$\sigma = 60, c = 36$, relevant	0.98	2.8	2.3	3.8	17	3.7	-12	-17
	$\sigma = 60, c = 36$, irrelevant	0.98	≤ 0.0	-0.5	-1.5	-3.7	-11	-26	-60
	$\sigma = 30, c = 1$, relevant	0.99	0.8	0.4	-0.3	-1.8	-9.9	-26	-59
	$\sigma = 30, c = 4$, relevant	0.98	1.2	1.3	1.9	3.4	-8.1	-24	-44
	$\sigma = 30, c = 36$, relevant	0.99	1.6	2.3	4.7	31	3.0	-13	> 0
	$\sigma = 60, c = 4$, relevant, 10% wrong	0.96	2.2	1.8	0.0	-0.2	-8.1	-24	-52
	$\sigma = 60, c = 4$, irrelevant, 10% wrong	0.96	≤ 0.0	-0.6	-1.1	-2.2	-9.6	-25	-58
	$\sigma = 60, c = 36$, relevant, 10% wrong	0.97	2.4	2.4	0.5	2.8	-3.6	-20	-33
	$\sigma = 60, c = 36$, irrelevant, 10% wrong	0.96	≤ 0.0	-0.7	-1.1	-2.3	-9.7	-25	-59
	$\sigma = 60, c = 4$, relevant, 50% wrong	0.95	2.3	1.7	0.1	-0.5	-8.3	-24	-53
	$\sigma = 60, c = 4$, irrelevant, 50% wrong	0.94	≤ 0.0	-0.5	-1.1	-2.3	-9.7	-25	-58
	$\sigma = 60, c = 36$, relevant, 50% wrong	0.95	2.8	2.3	0.7	1.2	-4.5	-21	-37
	$\sigma = 60, c = 36$, irrelevant, 50% wrong	0.95	≤ 0.0	-0.6	-1.1	-2.3	-9.7	-25	-59
Phi-4	$\sigma = 60, c = 1$, relevant	0.87	0.8	0.7	-0.2	-0.4	-1.0	-3.1	-35
	$\sigma = 60, c = 1$, irrelevant	0.86	≤ 0.0	≤ -1.0	≤ -3.0	-4.9	-6.1	-8.8	-40
	$\sigma = 60, c = 4$, relevant	0.85	5.8	4.1	2.4	2.1	0.7	-4.6	-33
	$\sigma = 60, c = 4$, irrelevant	0.84	≤ 0.0	≤ -1.0	≤ -3.0	-4.7	-6.2	-13	-42
	$\sigma = 60, c = 36$, relevant	0.86	15	7.6	4.7	5.6	8.5	> 32	> 0
	$\sigma = 60, c = 36$, irrelevant	0.84	≤ 0.0	≤ -1.0	≤ -3.0	-4.5	-6.4	-12	-42
	$\sigma = 30, c = 1$, relevant	0.87	1.1	0.8	-0.3	-0.3	-1.0	-1.8	-15
	$\sigma = 30, c = 4$, relevant	0.86	4.8	3.5	2.0	1.6	0.6	-1.9	-2.2
	$\sigma = 30, c = 36$, relevant	0.88	9.7	6.0	3.8	4.4	8.4	17	> 0
	$\sigma = 60, c = 4$, relevant, 10% wrong	0.87	6.0	5.5	5.3	8.3	15	> 32	> 0
	$\sigma = 60, c = 4$, irrelevant, 10% wrong	0.85	≤ 0.0	≤ -1.0	-2.1	-1.2	2.9	-3.2	> 0
	$\sigma = 60, c = 36$, relevant, 10% wrong	0.87	10	8.7	8.2	17	> 48	> 32	> 0
	$\sigma = 60, c = 36$, irrelevant, 10% wrong	0.86	≤ 0.0	≤ -1.0	-2.4	-1.7	1.3	-4.7	> 0
	$\sigma = 60, c = 4$, relevant, 50% wrong	0.87	6.0	6.4	7.7	17	> 48	> 32	> 0
	$\sigma = 60, c = 4$, irrelevant, 50% wrong	0.85	≤ 0.0	≤ -1.0	-1.3	1.5	16	> 32	> 0
	$\sigma = 60, c = 36$, relevant, 50% wrong	0.88	9.2	9.7	11	> 56	> 48	> 32	> 0
	$\sigma = 60, c = 36$, irrelevant, 50% wrong	0.86	≤ 0.0	≤ -1.0	-1.8	0.6	13	> 32	> 0
OLMo 2 32B	$\sigma = 60, c = 1$, relevant	0.90	1.9	1.8	2.0	2.3	5.8	> 32	> 0
	$\sigma = 60, c = 1$, irrelevant	0.89	≤ 0.0	-0.5	-0.2	-0.2	1.7	25	> 0
	$\sigma = 60, c = 4$, relevant	0.91	5.8	3.4	3.4	4.5	9.3	> 32	> 0
	$\sigma = 60, c = 4$, irrelevant	0.90	≤ 0.0	-0.7	-0.3	-0.5	0.3	17	> 0
	$\sigma = 60, c = 36$, relevant	0.91	11	4.6	5.3	6.8	15	> 32	> 0
	$\sigma = 60, c = 36$, irrelevant	0.90	≤ 0.0	≤ -1.0	-0.6	-0.5	-0.2	15	> 0
	$\sigma = 30, c = 1$, relevant	0.92	1.8	1.9	1.7	1.7	4.4	24	> 0
	$\sigma = 30, c = 4$, relevant	0.92	4.8	3.3	3.1	4.0	9.5	> 32	> 0
	$\sigma = 30, c = 36$, relevant	0.93	7.0	4.4	4.0	5.9	14	> 32	> 0
	$\sigma = 60, c = 4$, relevant, 10% wrong	0.96	6.0	4.6	3.8	4.5	5.8	23	> 0
	$\sigma = 60, c = 4$, irrelevant, 10% wrong	0.96	≤ 0.0	-0.9	0.0	0.0	-0.4	4.1	> 0
	$\sigma = 60, c = 36$, relevant, 10% wrong	0.96	12	5.9	6.0	6.3	8.9	31	> 0
	$\sigma = 60, c = 36$, irrelevant, 10% wrong	0.96	≤ 0.0	≤ -1.0	-0.3	-0.1	-0.8	0.3	> 0
	$\sigma = 60, c = 4$, relevant, 50% wrong	0.93	5.3	4.2	3.5	4.0	5.4	27	> 0
	$\sigma = 60, c = 4$, irrelevant, 50% wrong	0.92	≤ 0.0	-0.5	0.1	0.1	0.0	11	> 0
	$\sigma = 60, c = 36$, relevant, 50% wrong	0.93	7.8	5.4	4.9	5.6	7.9	> 32	> 0
	$\sigma = 60, c = 36$, irrelevant, 50% wrong	0.92	≤ 0.0	-0.8	-0.1	0.0	-0.3	7.8	> 0

Table 13. **ESS by description condition and demonstration count: written readout.** Results cover the three written-readout models in the full run without reasoning. Replies without a qualifying number are excluded. Columns and reference values follow Table 12.

Predictor	Description	number	$n = 1$	$n = 2$	$n = 4$	$n = 8$	$n = 16$	$n = 32$	$n = 64$
Qwen3.5 9B (written)	$\sigma = 60, c = 1$, relevant	1.00	0.2	0.3	−0.9	−2.8	−9.3	−24	−51
	$\sigma = 60, c = 1$, irrelevant	1.00	≤ 0.0	≤ -1.0	−2.7	−4.8	−9.7	−23	−49
	$\sigma = 60, c = 4$, relevant	1.00	3.9	2.7	1.3	−0.7	−6.2	−18	−44
	$\sigma = 60, c = 4$, irrelevant	1.00	≤ 0.0	≤ -1.0	−1.7	−3.8	−9.4	−24	−49
	$\sigma = 60, c = 36$, relevant	1.00	6.5	4.0	2.8	6.9	17	> 32	> 0
	$\sigma = 60, c = 36$, irrelevant	1.00	≤ 0.0	≤ -1.0	−2.8	−5.1	−10	−24	−51
	$\sigma = 30, c = 1$, relevant	1.00	≤ 0.0	0.2	−0.7	−2.5	−8.7	−22	−49
	$\sigma = 30, c = 4$, relevant	1.00	3.4	2.7	1.5	−0.3	−4.5	−14	−40
	$\sigma = 30, c = 36$, relevant	1.00	4.7	4.3	3.0	9.2	14	> 32	> 0
	$\sigma = 60, c = 4$, relevant, 10% wrong	1.00	3.8	−0.4	−0.7	−1.8	−8.5	−21	−47
	$\sigma = 60, c = 4$, irrelevant, 10% wrong	1.00	≤ 0.0	≤ -1.0	≤ -3.0	−4.9	−10	−24	−50
	$\sigma = 60, c = 36$, relevant, 10% wrong	1.00	5.3	0.4	0.6	−0.3	−3.3	−4.2	−6.9
	$\sigma = 60, c = 36$, irrelevant, 10% wrong	1.00	≤ 0.0	≤ -1.0	≤ -3.0	−5.2	−10	−25	−53
	$\sigma = 60, c = 4$, relevant, 50% wrong	1.00	4.5	−0.1	−0.4	−1.9	−9.0	−23	−50
	$\sigma = 60, c = 4$, irrelevant, 50% wrong	1.00	≤ 0.0	≤ -1.0	≤ -3.0	−6.2	−12	−26	−54
	$\sigma = 60, c = 36$, relevant, 50% wrong	1.00	7.3	0.8	1.0	−0.1	−4.0	−4.2	−12
	$\sigma = 60, c = 36$, irrelevant, 50% wrong	1.00	≤ 0.0	≤ -1.0	≤ -3.0	−7.0	−13	−27	−56
Gemma 4 12B (written)	$\sigma = 60, c = 1$, relevant	1.00	0.0	0.0	−0.1	−0.6	−3.2	−5.4	−21
	$\sigma = 60, c = 1$, irrelevant	1.00	≤ 0.0	−0.4	−1.3	−1.3	−4.4	−6.8	−24
	$\sigma = 60, c = 4$, relevant	1.00	0.3	0.5	1.4	1.1	−0.6	−2.7	−13
	$\sigma = 60, c = 4$, irrelevant	1.00	≤ 0.0	−0.4	−1.1	−1.2	−4.1	−6.9	−23
	$\sigma = 60, c = 36$, relevant	1.00	1.8	1.2	2.7	3.6	3.6	4.9	> 0
	$\sigma = 60, c = 36$, irrelevant	1.00	≤ 0.0	−0.8	−1.3	−1.3	−4.9	−7.9	−26
	$\sigma = 30, c = 1$, relevant	1.00	≤ 0.0	−0.2	−0.1	−0.5	−3.2	−7.9	−27
	$\sigma = 30, c = 4$, relevant	1.00	0.7	0.8	2.1	1.5	−0.2	−4.6	−18
	$\sigma = 30, c = 36$, relevant	1.00	4.0	1.7	3.4	3.7	2.4	−1.8	−8.9
	$\sigma = 60, c = 4$, relevant, 10% wrong	1.00	0.1	0.7	1.4	0.4	−1.0	−3.1	−7.3
	$\sigma = 60, c = 4$, irrelevant, 10% wrong	1.00	≤ 0.0	−0.2	−0.4	−0.5	−2.2	−5.4	−13
	$\sigma = 60, c = 36$, relevant, 10% wrong	1.00	0.6	1.3	2.3	1.6	0.7	−0.6	−1.3
	$\sigma = 60, c = 36$, irrelevant, 10% wrong	1.00	≤ 0.0	−0.2	−0.5	−0.5	−2.5	−5.8	−15
	$\sigma = 60, c = 4$, relevant, 50% wrong	1.00	1.0	0.6	1.7	1.0	−0.1	−1.7	−4.5
	$\sigma = 60, c = 4$, irrelevant, 50% wrong	1.00	≤ 0.0	−0.3	−0.4	−0.4	−1.3	−4.1	−9.6
	$\sigma = 60, c = 36$, relevant, 50% wrong	1.00	5.1	1.0	2.7	2.3	2.4	1.1	> 0
	$\sigma = 60, c = 36$, irrelevant, 50% wrong	1.00	≤ 0.0	−0.3	−0.6	−0.4	−1.9	−4.8	−13
Gemma 4 31B (written)	$\sigma = 60, c = 1$, relevant	1.00	0.1	0.0	−0.1	−0.5	−1.4	−5.1	−13
	$\sigma = 60, c = 1$, irrelevant	1.00	≤ 0.0	−0.1	−0.1	−0.5	−1.0	−4.5	−11
	$\sigma = 60, c = 4$, relevant	1.00	0.8	0.1	0.2	−0.5	−1.0	−4.3	−12
	$\sigma = 60, c = 4$, irrelevant	1.00	≤ 0.0	−0.2	−0.2	−0.4	−1.0	−4.4	−10
	$\sigma = 60, c = 36$, relevant	1.00	2.6	0.6	1.4	0.9	2.9	3.4	−2.9
	$\sigma = 60, c = 36$, irrelevant	1.00	≤ 0.0	≤ -1.0	−0.8	−0.6	−1.1	−2.7	−4.8
	$\sigma = 30, c = 1$, relevant	1.00	≤ 0.0	0.0	−0.1	−0.7	−1.2	−3.0	−8.1
	$\sigma = 30, c = 4$, relevant	1.00	0.5	0.0	0.1	−0.5	−1.3	−3.4	−8.3
	$\sigma = 30, c = 36$, relevant	1.00	4.3	1.5	2.1	2.3	3.3	3.3	−0.4
	$\sigma = 60, c = 4$, relevant, 10% wrong	1.00	0.1	0.0	0.0	−0.7	−1.9	−5.0	−15
	$\sigma = 60, c = 4$, irrelevant, 10% wrong	1.00	≤ 0.0	0.0	−0.1	−0.4	−1.2	−4.1	−9.5
	$\sigma = 60, c = 36$, relevant, 10% wrong	1.00	1.0	0.0	0.6	−0.3	−1.4	−4.8	−12
	$\sigma = 60, c = 36$, irrelevant, 10% wrong	1.00	≤ 0.0	0.0	−0.2	−0.4	−0.9	−3.0	−6.2
	$\sigma = 60, c = 4$, relevant, 50% wrong	1.00	0.2	0.0	0.0	−0.7	−1.7	−4.6	−17
	$\sigma = 60, c = 4$, irrelevant, 50% wrong	1.00	≤ 0.0	0.0	−0.2	−0.4	−1.2	−3.6	−12
	$\sigma = 60, c = 36$, relevant, 50% wrong	1.00	1.0	0.0	0.5	−0.3	−1.5	−4.5	−15
	$\sigma = 60, c = 36$, irrelevant, 50% wrong	1.00	≤ 0.0	0.0	−0.1	−0.4	−1.0	−2.9	−8.5

Table 14. **Description weights by demonstration count: distribution readout.** Results cover the prompt-based reference and the three distribution-readout models in the full run without reasoning. "Number" is defined as in Table 12; cells are italicized when the numeric response rate is below three quarters at the corresponding n.

Predictor	Description	number	$n=0$	$n=1$	$n=2$	$n=4$	$n=8$	$n=16$	$n=32$	$n=64$
Reference	$\sigma=60, c=1$, relevant		1.00	0.50	0.33	0.20	0.11	0.06	0.03	0.02
	$\sigma=60, c=1$, irrelevant		1.00	0.50	0.33	0.20	0.11	0.06	0.03	0.02
	$\sigma=60, c=4$, relevant		1.00	0.80	0.67	0.50	0.33	0.20	0.11	0.06
	$\sigma=60, c=4$, irrelevant		1.00	0.80	0.67	0.50	0.33	0.20	0.11	0.06
	$\sigma=60, c=36$, relevant		1.00	0.97	0.95	0.90	0.82	0.69	0.53	0.36
	$\sigma=60, c=36$, irrelevant		1.00	0.97	0.95	0.90	0.82	0.69	0.53	0.36
	$\sigma=30, c=1$, relevant		1.00	0.50	0.33	0.20	0.11	0.06	0.03	0.02
	$\sigma=30, c=4$, relevant		1.00	0.80	0.67	0.50	0.33	0.20	0.11	0.06
	$\sigma=30, c=36$, relevant		1.00	0.97	0.95	0.90	0.82	0.69	0.53	0.36
	$\sigma=60, c=4$, relevant, 10% wrong		0.90	0.71	0.59	0.45	0.30	0.19	0.10	0.05
	$\sigma=60, c=4$, irrelevant, 10% wrong		0.90	0.27	0.14	0.08	0.04	0.02	0.01	0.00
	$\sigma=60, c=36$, relevant, 10% wrong		0.90	0.84	0.84	0.82	0.77	0.65	0.51	0.34
	$\sigma=60, c=36$, irrelevant, 10% wrong		0.90	0.26	0.14	0.07	0.03	0.01	0.00	0.00
	$\sigma=60, c=4$, relevant, 50% wrong		0.50	0.46	0.39	0.32	0.22	0.14	0.08	0.04
	$\sigma=60, c=4$, irrelevant, 50% wrong		0.50	0.12	0.07	0.03	0.02	0.01	0.00	0.00
	$\sigma=60, c=36$, relevant, 50% wrong		0.50	0.55	0.58	0.60	0.59	0.53	0.42	0.29
	$\sigma=60, c=36$, irrelevant, 50% wrong		0.50	0.12	0.06	0.03	0.01	0.00	0.00	0.00
gpt-oss-20b	$\sigma=60, c=1$, relevant	0.98	0.94	0.53	0.29	0.23	0.16	0.12	0.12	0.23
	$\sigma=60, c=1$, irrelevant	0.98	0.93	0.59	0.25	0.18	0.09	0.05	0.04	0.10
	$\sigma=60, c=4$, relevant	0.98	0.92	0.48	0.32	0.26	0.21	0.19	0.18	0.31
	$\sigma=60, c=4$, irrelevant	0.97	0.92	0.57	0.25	0.16	0.08	0.04	0.04	0.09
	$\sigma=60, c=36$, relevant	0.98	0.94	0.49	0.35	0.30	0.25	0.23	0.22	0.35
	$\sigma=60, c=36$, irrelevant	0.98	0.95	0.60	0.25	0.15	0.07	0.04	0.04	0.09
	$\sigma=30, c=1$, relevant	0.99	0.98	0.43	0.23	0.17	0.12	0.09	0.11	0.21
	$\sigma=30, c=4$, relevant	0.98	0.98	0.38	0.28	0.23	0.19	0.18	0.19	0.31
	$\sigma=30, c=36$, relevant	0.99	0.99	0.37	0.32	0.27	0.23	0.22	0.24	0.37
	$\sigma=60, c=4$, relevant, 10% wrong	0.96	0.91	0.45	0.33	0.15	0.13	0.12	0.13	0.22
	$\sigma=60, c=4$, irrelevant, 10% wrong	0.96	0.91	0.55	0.34	0.11	0.05	0.02	0.02	0.04
	$\sigma=60, c=36$, relevant, 10% wrong	0.97	0.92	0.44	0.34	0.17	0.15	0.15	0.16	0.25
	$\sigma=60, c=36$, irrelevant, 10% wrong	0.96	0.92	0.53	0.34	0.10	0.04	0.02	0.02	0.04
	$\sigma=60, c=4$, relevant, 50% wrong	0.95	0.79	0.46	0.32	0.16	0.12	0.12	0.12	0.20
	$\sigma=60, c=4$, irrelevant, 50% wrong	0.94	0.79	0.53	0.32	0.11	0.04	0.02	0.01	0.04
	$\sigma=60, c=36$, relevant, 50% wrong	0.95	0.83	0.46	0.32	0.17	0.14	0.15	0.15	0.23
	$\sigma=60, c=36$, irrelevant, 50% wrong	0.95	0.83	0.53	0.31	0.11	0.04	0.02	0.02	0.04
Phi-4	$\sigma=60, c=1$, relevant	0.87	*0.97*	0.82	0.62	0.43	0.28	0.18	0.13	0.12
	$\sigma=60, c=1$, irrelevant	0.86	*0.97*	0.83	0.64	0.42	0.22	0.11	0.06	0.05
	$\sigma=60, c=4$, relevant	0.85	*0.98*	0.76	0.55	0.37	0.26	0.19	0.15	0.16
	$\sigma=60, c=4$, irrelevant	0.84	*0.98*	0.79	0.56	0.36	0.18	0.09	0.06	0.05
	$\sigma=60, c=36$, relevant	0.86	*0.99*	0.74	0.53	0.38	0.29	0.23	0.20	0.20
	$\sigma=60, c=36$, irrelevant	0.84	*0.99*	0.78	0.53	0.33	0.16	0.09	0.06	0.04
	$\sigma=30, c=1$, relevant	0.87	*0.99*	0.77	0.58	0.38	0.22	0.13	0.09	0.09
	$\sigma=30, c=4$, relevant	0.86	*1.00*	0.69	0.51	0.34	0.23	0.16	0.13	0.13
	$\sigma=30, c=36$, relevant	0.88	*1.00*	0.65	0.48	0.32	0.23	0.18	0.17	0.16
	$\sigma=60, c=4$, relevant, 10% wrong	0.87	*0.94*	0.62	0.46	0.30	0.21	0.15	0.13	0.11
	$\sigma=60, c=4$, irrelevant, 10% wrong	0.85	*0.95*	0.62	0.44	0.26	0.13	0.06	0.04	0.03
	$\sigma=60, c=36$, relevant, 10% wrong	0.87	*0.95*	0.61	0.46	0.33	0.25	0.20	0.18	0.16
	$\sigma=60, c=36$, irrelevant, 10% wrong	0.86	*0.96*	0.61	0.43	0.26	0.13	0.06	0.04	0.03
	$\sigma=60, c=4$, relevant, 50% wrong	0.87	*0.92*	0.57	0.43	0.29	0.20	0.15	0.13	0.11
	$\sigma=60, c=4$, irrelevant, 50% wrong	0.85	*0.92*	0.57	0.42	0.24	0.12	0.06	0.04	0.03
	$\sigma=60, c=36$, relevant, 50% wrong	0.88	*0.93*	0.56	0.44	0.33	0.24	0.21	0.19	0.17
	$\sigma=60, c=36$, irrelevant, 50% wrong	0.86	*0.94*	0.57	0.41	0.24	0.12	0.06	0.04	0.03
OLMo 2 32B	$\sigma=60, c=1$, relevant	0.90	*0.99*	0.72	0.33	0.25	0.16	0.09	0.05	0.05
	$\sigma=60, c=1$, irrelevant	0.89	*0.99*	0.77	0.29	0.16	0.09	0.05	0.02	0.02
	$\sigma=60, c=4$, relevant	0.91	*1.00*	0.70	0.37	0.30	0.21	0.14	0.08	0.07
	$\sigma=60, c=4$, irrelevant	0.90	*1.00*	0.75	0.29	0.15	0.08	0.04	0.02	0.01
	$\sigma=60, c=36$, relevant	0.91	*1.00*	0.70	0.40	0.34	0.26	0.19	0.12	0.10
	$\sigma=60, c=36$, irrelevant	0.90	*1.00*	0.76	0.29	0.17	0.08	0.04	0.02	0.02
	$\sigma=30, c=1$, relevant	0.92	*0.99*	0.69	0.32	0.23	0.14	0.09	0.04	0.04
	$\sigma=30, c=4$, relevant	0.92	*1.00*	0.65	0.34	0.27	0.19	0.13	0.06	0.06
	$\sigma=30, c=36$, relevant	0.93	*1.00*	0.64	0.36	0.32	0.23	0.16	0.08	0.09
	$\sigma=60, c=4$, relevant, 10% wrong	0.96	1.01	0.72	0.42	0.29	0.18	0.11	0.09	0.05
	$\sigma=60, c=4$, irrelevant, 10% wrong	0.96	1.01	0.81	0.32	0.14	0.06	0.03	0.02	0.01
	$\sigma=60, c=36$, relevant, 10% wrong	0.96	1.01	0.71	0.44	0.33	0.23	0.15	0.13	0.07
	$\sigma=60, c=36$, irrelevant, 10% wrong	0.96	1.01	0.82	0.33	0.15	0.07	0.03	0.02	0.01
	$\sigma=60, c=4$, relevant, 50% wrong	0.93	1.02	0.65	0.39	0.27	0.16	0.09	0.07	0.04
	$\sigma=60, c=4$, irrelevant, 50% wrong	0.92	1.01	0.71	0.29	0.13	0.06	0.02	0.01	0.01
	$\sigma=60, c=36$, relevant, 50% wrong	0.93	1.02	0.64	0.42	0.31	0.21	0.13	0.11	0.06
	$\sigma=60, c=36$, irrelevant, 50% wrong	0.92	1.02	0.72	0.30	0.14	0.06	0.03	0.02	0.01

D. Further Related Work

Descriptions as substitutes for examples. Beyond in-context learning, an instruction can replace up to 100 execution examples in the fine-tuning settings studied by Cook et al. (2026). Wenliang et al. (2026) compare optimized prompts with demonstrations in terms of their length and effect. These results motivate measuring descriptions relative to examples, while differing from our predictive-risk comparison within a fixed task distribution and learner.

Text and probabilistic prediction. Text supplied alongside numbers can change numerical predictions (Requeima et al., 2024). Related work studies how closely language-model belief updates follow Bayesian inference and how training can improve probabilistic reasoning (Qiu et al., 2026). He et al. (2025) examine Bayesian consistency during iterative reasoning on open-ended questions. Our experiments use a controlled numerical task with explicit uncertainty so that description precision, relevance, and prediction weights can be compared with a specified probabilistic reference.